

Final Project for SW Engineering Class CSC 648-848 Spring 2025

Team 08

Project Name: GatorTrade

Team Lead: Jonathan Carbajal - Jcarbajal@sfsu.edu
Backend Lead: Christian Kurt Balais - cbalais@sfsu.edu
Frontend Lead: Tung-Ying Lee - nlee@sfsu.edu
Frontend Lead 2: Chia-Tung Liu - cliu26@sfsu.edu
Github Master: Luis Moreno - lmoreno10@sfsu.edu

URL: <http://18.117.162.109/>

Date: 05/20/2025

I. Product Summary:

Product name: Gator Trade

Major Committed Functions

- Browsing Items: Anyone can browse all items, including categories, prices, images, and descriptions.
- Searching & Filtering: Search by keyword, filter by category, and set a price range to narrow down results.
- Viewing Item Details: View full item info like images, description, price, and posting time.
- Student Registration: Only users with a valid SFSU email can register.
- Login Access: Registered users can securely log in to access full features.
- Posting Items for Sale: Users can create listings with images and details (admin-reviewed before posting).
- Managing Listings: Users can delete their listings or mark them as sold.
- In-Site Messaging: Buyers and sellers communicate through a private chat system.
- Seller Inbox & Notifications: Sellers are notified via their SFSU email when a buyer contacts them.
- Profile Management: Users can manage personal details like name and profile picture.
- Safe Meetup Locations: Predefined campus locations are suggested for safe exchanges, sent via message template.

GatorTrade is exclusively for the SFSU community, requiring all users to register using their official university email. This tight-knit access ensures a safer, scam-resistant platform for students and faculty. It is student-focused with dedicated categories for textbooks, school supplies, and dorm needs, streamlining buying and selling on campus. Admin moderation and safe meet-up location tools further enhance safety and relevance.

Product URL: <http://18.117.162.109/>

II. Milestone Documents:

Full History Table:

Version:	Date Submitted:	Date Revised:
M1	03-12-2025	03-21-2025
M2	03-25-2025	04-03-2025
M3	04-26-2025	N/A
M4	05-20-2025	N/A
M5	05-20-2025	N/A

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Milestone 1

History Table:

Version:	Date Submitted:	Date Revised:
1	03-12-2025	03-21-2025

I. Executive Summary

While popular, platforms like Facebook Marketplace, Craigslist, OfferUp, and eBay have challenges that impact SFSU students daily. Buying and selling essentials like electronics, furniture, and textbooks can be frustrating due to scams, scalpers, and location-based limitations. Additionally, these platforms prioritize corporate interests, leading to issues like data privacy concerns, intrusive ads, and irrelevant listings. Motivated by these challenges, my team and I created GatorTrade, a dedicated, secure, and student-exclusive platform designed to ensure every buyer and seller is a member of the SFSU community.

Since GatorTrade is a secure marketplace where only SFSU students and faculty can participate, all users must have university email accounts. This eliminates the risks of anonymous buyers and sellers to ensure a safe and secure environment. While it is possible for individuals within the SFSU community to engage in misconduct, such as scams, harassment, or theft, our platform prioritizes security by adding an admin system that monitors and moderates listings and interactions to maintain a safe and trustworthy marketplace. To further streamline transactions, the platform includes additional features such as **integrated chat for direct communication**, suggested **on-campus meetup locations** for safe exchanges, **item tracking** to monitor availability and status, and **a profile rating system** that helps users build trust within the community. These tools not only make transactions more efficient but also reinforce the security and reliability of the marketplace. With its unique student-centric design, GatorTrade will focus on those students in need, unlike other marketplaces that cater to a broad audience. Different dedicated sections for textbooks, school supplies, and dorm essentials will be boosted to make navigating for these student materials easier.

Our team is dedicated to creating this website for our community with focus and passion. As senior-year computer science majors, we want to showcase the skills we have acquired throughout our studies and apply them to a project that makes a meaningful impact.

II. Personae

Persona 1: James Tran- The Casual Seller

Name: James Tran

Age: 28

Profile: SFSU Graduate Student in Computer Science

Lives in: Daly City

Goals:

- Sell used items quickly and efficiently within the SFSU area
- Avoid dealing with shipping or returns
- Get a fair price without long negotiation

Skills:

- Understands how to attract local buyers
- Uses secure and verified buyer features to avoid scams
- Prefers a simple and efficient listing process

Pain Points:

- No-shows or last-minute cancellations from buyers
- Managing multiple inquiries and lowball offers
- Finding a safe and convenient meeting point near SFSU or BART Station

Use Case Connection:

- **Listing Item:** James, a senior, wants to sell his old gaming chair before leaving SFSU. This directly connects to James Tran's need for quick and efficient sales and his skill in using tools like the proximity radar feature to find local buyers.

Persona 2: Emma Chen - The Frugal Student

Name: Emma Chen

Age: 20

Profile: SFSU Undergraduate Student

Lives in: Sunset District

Goals:

- Find affordable furniture, textbooks, and essentials for her apartment

- Buy locally to avoid shipping fees
- Meet sellers in safe, accessible locations near campus

Skills:

- Budget-conscious shopping with a preference for verified sellers
- Comfortable with in-person pick-ups
- Relies on third-party evaluations for item quality

Pain Points:

- Making sure sellers are trustworthy and items are as described
- Limited transportation options for larger purchases
- Avoiding scams and overpricing

Use Case Connection:

- **Browsing for Items:** Emma is actively searching for a bike using filters to stay within her budget. This aligns with her persona as a frugal student who values affordability and safe in-app communication.

Persona 3: Michael Rivera - The Small Business Reseller

Name: Michael Rivera

Age: 50

Profile: SFSU Faculty Member, Art Department

Lives in: Parkmerced

Goals:

- Find local buyers for second-hand and upcycled goods
- Avoid expensive e-commerce platforms and shipping fees
- Sell high-quality vintage goods quickly
- Attract SFSU students who appreciate sustainable fashion and unique home decor

Skills:

- Curating rare vintage items that resonate with younger buyers
- Experience with resale trends and student preferences
- Prefers platforms with no marketplace commissions and trusted buyer verification

Pain Points:

- Managing multiple sales platforms

- Ensuring safe transactions without fraud
- Competing with larger online stores

Use Case Connection:

- **Meeting Up and Completing the Transaction:** Michael meets a student buyer at the SFSU library to complete a safe, in-person transaction. After the deal, the buyer tags him as a trusted seller, aligning with Michael's desire for verified transactions and building trust on campus.

III. High-Level Use Case

A. User registration and login

1. A classmate tells **Emma Chen** about the student marketplace app after she mentions needing affordable items for her apartment. Curious, Emma decides to check it out to browse for a bike and some furniture. She downloads the app and registers using her SFSU student email. The system verifies her identity to ensure only legitimate students can join. Once validated, Emma accesses the main dashboard where she can browse local listings, message sellers, and create a wish list for essential items she's looking for.

B. Listing Item

1. **James Tran**, a graduate student preparing to move out of Daly City, wants to sell his old gaming chair. He opens the marketplace app, heads to the "Sell" section, and starts creating a listing. He selects "Furniture" as the category, sets a reasonable price, adds a description emphasizing the chair's great condition, and uploads clear photos. To ensure a fast sale within the SFSU community, he activates the proximity radar to notify students nearby. After reviewing his listing, James publishes it, making it visible to verified students searching for furniture in the area.

C. Browsing For Items

1. **Emma Chen** is on the hunt for a used bike to make her commute around campus easier. She logs into the app and searches for "bikes near SFSU." Using filters, she narrows down listings to those within her budget and preferred distance from Sunset. She finds a promising bike, checks the seller's "trusted" rating, and reads through the listing details. To learn more, Emma messages the seller through the app's secure chat, keeping her personal information private while arranging a potential meet-up.

D. Meeting Up and Completing the Transaction

1. **Michael Rivera**, an SFSU faculty member and part-time reseller, has listed a vintage laptop stand for sale. After receiving a message from a student buyer, they agree to meet at the SFSU library to complete the transaction. The app helps Michael coordinate by providing the buyer with a general location pin, maintaining privacy. Upon inspecting the item in person, the student is satisfied and completes the purchase. Afterward, the buyer rates Michael as a "trusted seller," boosting his credibility within the app's community.

E. Reporting Listing

1. While browsing for textbooks, **James Tran** comes across a suspicious listing offering a high-demand textbook at an unusually low price with vague item details. Concerned about a potential scam targeting students, James clicks "Report," selects "Possible Fraud" from the dropdown options, and adds a short note explaining the issue. The report is sent to the app's moderation team, who will investigate and take necessary action to maintain a secure and trusted marketplace for SFSU students.

IV. Data Glossary/Description

A. User (Buyer/Seller/Admin)

1. Unique ID
2. First Name
3. Last Name
4. Email
5. Password
6. Role

B. Listings

1. Unique ID
2. User ID
3. Title
4. Description
5. Price
6. Category
7. Image URL
8. Status

C. Messages

1. Unique ID
2. Receiver ID
3. Sender ID
4. Listing ID
5. Content
6. Timestamp

D. Admin Review Logs

1. Unique ID
2. Admin ID
3. Listing ID
4. Decision
5. Moderator Notes
6. Timestamp (for reviewing listings)

V. High-Level Functional Requirements

1. **Browsing items-** Unregistered users shall view all listed items, including categories, prices, images, and descriptions.
2. **Searching and filtering items-** Unregistered users shall be able to access keyword search, category filtering, and price range filtering
3. **Viewing item details-** Shall allow users to click on an item to view the full description, images, price, and time of posting.
4. **User registration -** Unregistered users shall sign up using a valid email/university email only.
5. **Posting items for sale-** Users shall be able to fill in item details, upload images, and submit items for admin approval before they go live.
6. **Managing posted items-** Sellers shall be able to view, edit, or delete their listings from the “My Listings” section.
7. **In-site messaging system-** Buyers shall be able to contact sellers using an in-platform messaging system, while sellers shall be able to view received messages from the dashboard.
8. **Favoriting/tracking items-** Users shall be able to add items to the “Favorites” list for easy access later.
9. **Managing user profile-** Users shall be able to update their information in the future, except for their email.
10. **Reporting inappropriate items-** Users shall be able to report listings for violations such as fraudulent activity or inappropriate content.
11. **Approving/rejecting items –** Admins shall review all newly posted items before they go live to ensure compliance with platform policies.
12. **Removing inappropriate listings –** Admins shall be able to delete listings that violate marketplace rules and notify the seller.
13. **Banning users –** Admins shall have the authority to suspend or permanently ban users who repeatedly violate marketplace policies.

14. **Managing system announcements** – Admins shall be able to post announcements about maintenance schedules or platform policy updates.
15. **University email verification only**- Users shall be required to register and participate in transactions using a valid university email.
16. **School Specific Listings**- There shall be a dedicated category for college students to trade school-related items (e.g., textbooks, school supplies).
17. **University event promotions**- The platform shall allow student clubs and organizations to post event-related listings (e.g., graduation furniture sales).
18. **Verified review system for students**- Buyers and sellers shall be able to leave reviews only after completing a transaction, and these reviews shall affect the user's credibility.
19. **Lost and found section for university students**- Users shall be able to post lost items or list found belongings to help reunite owners with their possessions.
20. **On-campus meetup locations**- The platform shall suggest safe meetup locations on university campuses (e.g., library, student center).
21. **User Login** - Users shall be able to log in and access features previously locked to unregistered users.

VI. Non-Functional Requirement

1. The application shall be developed, tested, and deployed using tools and cloud servers approved by Class CTO and as agreed in M0.
2. The application shall be optimized for standard desktop/laptop browsers; e.g., it must render correctly on the two latest versions of two major browsers.
3. All or selected application functions shall render well on mobile devices (no native app to be developed).
4. Posting of sales information and messaging to sellers shall be limited only to SFSU students.
5. Critical data shall be stored in the database on the team's deployment server.

6. No more than 50 concurrent users shall be accessing the application at any time.
7. The privacy of users shall be protected.
8. The language used shall be English (no localization needed).
9. The application shall be easy to use and intuitive.
10. The application shall follow established architecture patterns.
11. The application code and its repository shall be easy to inspect and maintain.
12. Google Analytics shall be used.
13. No e-mail clients or chat services shall be allowed. Interested users can only message sellers via in-site messaging. One round of messaging (from user to seller) is enough for this application.
14. Pay functionality, if any (e.g., paying for goods and services), shall not be implemented nor simulated in UI.
15. Site security: Basic best practices shall be applied (as covered in the class) for main data items.
16. Media formats shall be standard as used in the market today.
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools.
18. The application UI (WWW and mobile) shall prominently display the exact text on all pages: "SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only" at the top of the WWW page Nav bar. (It is important to not confuse this with a real application). You have to use this exact text without any editing.

VII. Competitive Analysis

Feature	Facebook Marketplace	OfferUp	Craigslist	Our app GatorTrade (Planned)
User Verification	Facebook account (no strict verification)	Phone/email-based, no strict ID check	No verification (completely anonymous)	SFSU Student Email & ID Verification
Messaging System	Messenger (requires Facebook account)	In-app chat system	No built-in chat (email relay or direct contact)	In-app chat system (private & secure)
Proximity-Based Search	Yes, based on user location	Yes, with adjustable distance	Yes, but no built-in map/radius search	Yes, it is a rough location estimate (proximity radar)
Target Audience	General public	General public	General public	SFSU Students Only (Verified)
Trust & Safety	Scams & fake accounts exist	Some fraud detection still faces scams	High scam risk due to lack of verification	More secure due to student verification & reporting system
Transaction Type	Local meetups only	Local meetups + shipping available	Local meetups only	Local meetups, campus-focused transactions

Compared to Craigslist, OfferUp, and Facebook Marketplace, the SFSU Student Marketplace stands out as an exclusive, student-only network that guarantees more security and confidence. This network requires SFSU student email and ID verification to prevent fraud, in contrast to Craigslist, which permits completely anonymous ads, or OfferUp and Facebook Marketplace, which are troubled with fraudulent identities and frauds. Furthermore, Facebook needs Messenger, whereas Craigslist does not have an integrated chat system; our platform will have a safe, in-app messaging system for smooth collaboration. Lastly, our proximity radar strikes a balance between student safety and convenience by providing approximate location estimations without disclosing precise locations. Our marketplace provides SFSU students with a

safer, more dependable option that is suited to their needs by emphasizing a reputable campus community

VIII. System Architecture and Tech Used

- A.** Server Host: AWS EC2
- B.** Operating System: Ubuntu 22.04 LTS
- C.** Database: MySQL version 8
- D.** Web Server: NGINX 1.27.4
- E.** Front-End Framework: React
- F.** Server-Side Language: JavaScript
- G.** Web Framework: Express.js
- H.** Supported Browsers: Chrome, Safari
- I.** IDE: Visual Studio Code
- J.** SSL Cert: Let's Encrypt (via Certbot)
- K.** Additional Software: PM2, Git Bash, Node JS

IX. Use of GenAI Tools

- A. Brainstorming Ideas** – Use ChatGPT to suggest potential app features, user scenarios, or improvements based on competitors. Use ChatGPT to brainstorm a website name.
- B. Drafting Content** – Generate drafts for executive summaries, user personas, and functional requirements.
- C. Use Case Suggestions** – Ask ChatGPT to suggest possible use cases based on your app's purpose.
- D. Code Assistance** – Get help with coding logic, debugging, or understanding frameworks.
- E. Competitive Analysis** – Use ChatGPT to quickly summarize competitor features.
- F. Writing & Formatting Help** - Use ChatGPT to refine grammar, structure, and clarity in reports.
- G. Rating(out of 5):**
 - 1. **Brainstorming Ideas** – 3
 - 2. **Drafting Content** – 2
 - 3. **Use Case Suggestions** – 2
 - 4. **Code Assistance** – 3
 - 5. **Competitive Analysis** – 2
 - 6. **Writing & Formatting Help** – 1

X. Team and Roles

- A. Team Lead:** Jonathan Carbajal - Jcarbajal@sfsu.edu
- B. Backend Lead:** Kurt Balais - cbalais@sfsu.edu
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XI. Team Lead Checklist

- A.** So far, all team members are fully engaged and attending team sessions when required: **DONE**
- B.** The team found a time slot to meet outside of class: **DONE**
- C.** The team is ready and able to use the chosen back and front-end frames, and those who need to learn are working on learning/practicing: **DONE**
- D.** The team reviewed class slides on requirements and use cases before drafting minestrone 1: **DONE**
- E.** The team reviewed non-functional requirements from the “How to start...” document and developed Milestone 1 consistently: **DONE**
- F.** Team lead checked the Milestone 1 document for quality, completeness, formatting, and compliance with instructions before submission: **DONE**
- G.** The team lead ensured all team members read the final M1 and agreed/understood it before submission: **DONE**
- H.** The team shared and discussed their experience with GenAI tools among themselves: **DONE**
- I.** Github organized as discussed in class: **ON TRACK**

- END OF MILESTONE 1 DOCUMENT

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Milestone 2 Part 1

Date: 03/25/2025

History Table:

Version:	Date Submitted:	Date Revised:
M2	03-25-2025	04-03-2025

I. Executive Summary

While popular, platforms like Facebook Marketplace, Craigslist, OfferUp, and eBay have challenges that impact SFSU students daily. Buying and selling essentials like electronics, furniture, and textbooks can be frustrating due to scams, scalpers, and location-based limitations. Additionally, these platforms prioritize corporate interests, leading to issues like data privacy concerns, intrusive ads, and irrelevant listings. Motivated by these challenges, my team and I created GatorTrade, a dedicated, secure, and student-exclusive platform designed to ensure every buyer and seller is a member of the SFSU community.

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Our team is dedicated to creating this website for our community with focus and passion. As senior-year computer science majors, we want to showcase the skills we have acquired throughout our studies and apply them to a project that makes a meaningful impact.

II. List of Main Data Items and Entities

1. *User (Buyer/Seller/Admin)*

Represents a registered individual on the platform.

- **User_id** – Unique identifier for the user (**mandatory**)
- **First_name** – User's first name (**mandatory**)
- **Last_name** – User's last name (**mandatory**)
- **Email** – User's unique email address (**mandatory**)
- **Password_hash** – Encrypted password for account login (**mandatory**)
- **Role** – The user's role being Admin or non-admin (e.g., Admin or Regular User) (**mandatory**)
- **Created_at** – Date the account was created (**mandatory**)

2. *Listing*

Represents an item posted by a seller.

- **Listing_id** – Unique identifier for the listing (**mandatory**)
- **user_id** (Seller) – Reference to the seller's user ID (**mandatory**)
- **Title** – Title of the item (**mandatory**)
- **Description** – Full description of the item (**mandatory**)
- **Price** – Price set by the seller (**mandatory**)
- **Category** – Category the item belongs to (e.g., Furniture, Textbooks) (**mandatory**)
- **image_url(s)** – One or more image links (**mandatory**)
- **Thumbnail** – Lower resolution of the item's first image (**mandatory**)
- **Status** – Listing status (e.g., Pending Approval, Active, Sold) (**mandatory**)
- **Created_at** – When the item was posted (**mandatory**)
- **Location** (proximity data or campus area) – Campus area or proximity-based info (**mandatory**)

3. *Message*

Represents a message sent between users.

- **Message_id** – Unique identifier for the message (**mandatory**)
- **Receiver_id** – Sender's user ID (**mandatory**)
- **Sender_id** – Receiver's user ID (**mandatory**)
- **Listing_id** – Reference to associated listing (**mandatory**)
- **Content** – Text of the message (**mandatory**)
- **Timestamp** – When the message was sent (**mandatory**)

4. *Admin Review Log*

Captures moderation actions performed by Admins on listings.

- **Admin_review_log_id** – Unique review log ID (**mandatory**)
- **Admin_id** – Admin who reviewed the listing (**mandatory**)
- **Listing_id** – Listing under review (**mandatory**)
- **Decision** – Outcome of the review (Approved / Rejected) (**mandatory**)

- **Moderator_Notes** – Notes added by the moderator (**optional**)
- **Reviewed_at** – When the review was completed (**mandatory**)

5. *Review*

Represents buyer feedback on a seller after a transaction.

- **Review_id** – Unique identifier for the review (**Mandatory**)
- **Reviewer_id** – User ID of the reviewer (Buyer) (**Mandatory**)
- **Reviewee_id** – User ID of the reviewee (Seller) (**Mandatory**)
- **Listing_id** – Related listing being reviewed (**Mandatory**)
- **Rating** – Numeric rating score (e.g., 1–5 stars) (**Mandatory**)
- **Comment** – Textual feedback from buyer (**Optional**)
- **Created_at** – When the review was submitted (**Mandatory**)

6. *Favorite/Watchlist*

Tracks listings saved or favorited by users.

- **Favorite_id** – Unique identifier for the saved item (**Mandatory**)
- **User_id** – Reference to the saving user's ID (**Mandatory**)
- **Listing_id** – Listing that was saved/favorited (**Mandatory**)
- **Timestamp** – Timestamp of when it was added to watchlist (**Mandatory**)

7. *Reports*

Used by users to report inappropriate or fraudulent listings.

- **Report_id** – Unique identifier for the report (**Mandatory**)
- **Reporter_id** – User ID of the person reporting (**Mandatory**)
- **Listing_id** – Listing being reported (**Mandatory**)
- **Report_type** – Type/category of report (e.g., Fraud, Inappropriate Content) (**Mandatory**)
- **Report_description** – User explanation of the issue (**Optional**)
- **Timestamp** – When the report was submitted (**Mandatory**)
- **Review_status** – Status of report review (Pending, Reviewed, Action Taken) (**Mandatory**)

8. *System Announcement*

Admin-generated announcements visible to users.

- **Announcement_id** – Unique identifier for the announcement (**Mandatory**)
- **Admin_id** – Admin user who posted the announcement (**Mandatory**)
- **Title** – Title of the announcement (**Mandatory**)
- **Content** – Full message content (**Mandatory**)
- **Timestamp** – When the announcement was posted (**Mandatory**)
- **Status** – Whether the announcement is still active (Active/Expired) (**Mandatory**)

9. *Lost and Found*

Platform section for posting lost or found items.

- **Post_id** – Unique identifier for the post (**Mandatory**)

- **Poster_id** – User who posted the lost/found item (**Mandatory**)
- **Title** – Brief title of the post (**Mandatory**)
- **Description** – Description of the lost/found item (**Mandatory**)
- **Category** – Whether the item is lost or found (Lost or Found) (**Mandatory**)
- **Location_description** – Description of where it was found/lost (**Mandatory**)
- **Contact_method** – How to contact the poster (e.g., email, phone) (**Mandatory**)
- **Image_url** – Photo of the item (**Optional**)
- **Timestamp** – When the item was posted (**Mandatory**)

10. Event Promotion

For promoting campus events or sales by students or orgs.

- **Event_id** – Unique identifier for the event post (**Mandatory**)
- **Poster_id** – User or organization who posted the event (**Mandatory**)
- **Title** – Title of the event (**Mandatory**)
- **Description** – Description and details of the event (**Mandatory**)
- **Event_date** – Date the event is happening (**Mandatory**)
- **Category** – Type of event (e.g., Graduation Sale) (**Mandatory**)
- **Image_url** – Image or flyer for the event (**Optional**)
- **Timestamp** – When the event post was created (**Mandatory**)

III. Functional Requirements

Priority 1 – Must Have

Unregistered Users

- **1 Browsing items** – Unregistered users shall view all listed items, including categories, prices, images, and descriptions.
- **2 Searching and filtering items** – Unregistered users shall be able to access keyword search, category filtering, and price range filtering.
- **3 Viewing item details** – Shall allow users to click on an item to view the full description, images, price, and time of posting.
- **4 User registration** – Unregistered users shall sign up using a valid email/university email.
- **15 University email verification only** – Users shall be required to register and participate in transactions using a valid university email.
- **21 User Login** - Users shall be able to log in and access features previously locked to unregistered users.

Registered Users

- **5 Posting items for sale** – Users shall be able to fill in item details, upload images, and submit items for admin approval before they go live.
- **6 Managing posted items** – Sellers shall be able to view, edit, or delete their listings from the “My Listings” section.
- **7 In-site messaging system** – Buyers shall be able to contact sellers using an in-platform messaging system, while sellers shall be able to view received messages from the dashboard.

Admin

- **^11 Approving/rejecting items** – Admins shall review all newly posted items before they go live to ensure compliance with platform policies.

Priority 2 – Desired

Registered Users

- **8. Favoriting/tracking items** – Users shall be able to add items to the “Favorites” list for easy access later.
- **9. Managing user profile** – Users shall be able to update their information in the future, except for their email.
- **10. Reporting inappropriate items** – Users shall be able to report listings for violations such as fraudulent activity or inappropriate content.
- **13 Banning users** – Admins shall have the authority to suspend or permanently ban users who repeatedly violate marketplace policies.
- **20. On-campus meetup locations** – The platform shall suggest safe meetup locations on university campuses (e.g., library, student center).

Admin

- **12. Removing inappropriate listings** – Admins shall be able to delete listings that violate marketplace rules and notify the seller.
- **14. Managing system announcements** – Admins shall be able to post announcements about maintenance schedules or platform policy updates.

Priority 3 – Opportunistic

Registered Users

- **16. Miscellaneous Listing**– There shall be a dedicated category for random items that may not be school-related.
- **18. Verified review system for students** – Buyers and sellers shall be able to leave reviews only after completing a transaction, and these reviews shall affect the user’s credibility.
- **19. Lost and found section for university students** – Users shall be able to post lost items or list found belongings to help reunite owners with their possessions.

Other Users (Clubs/Organizations)

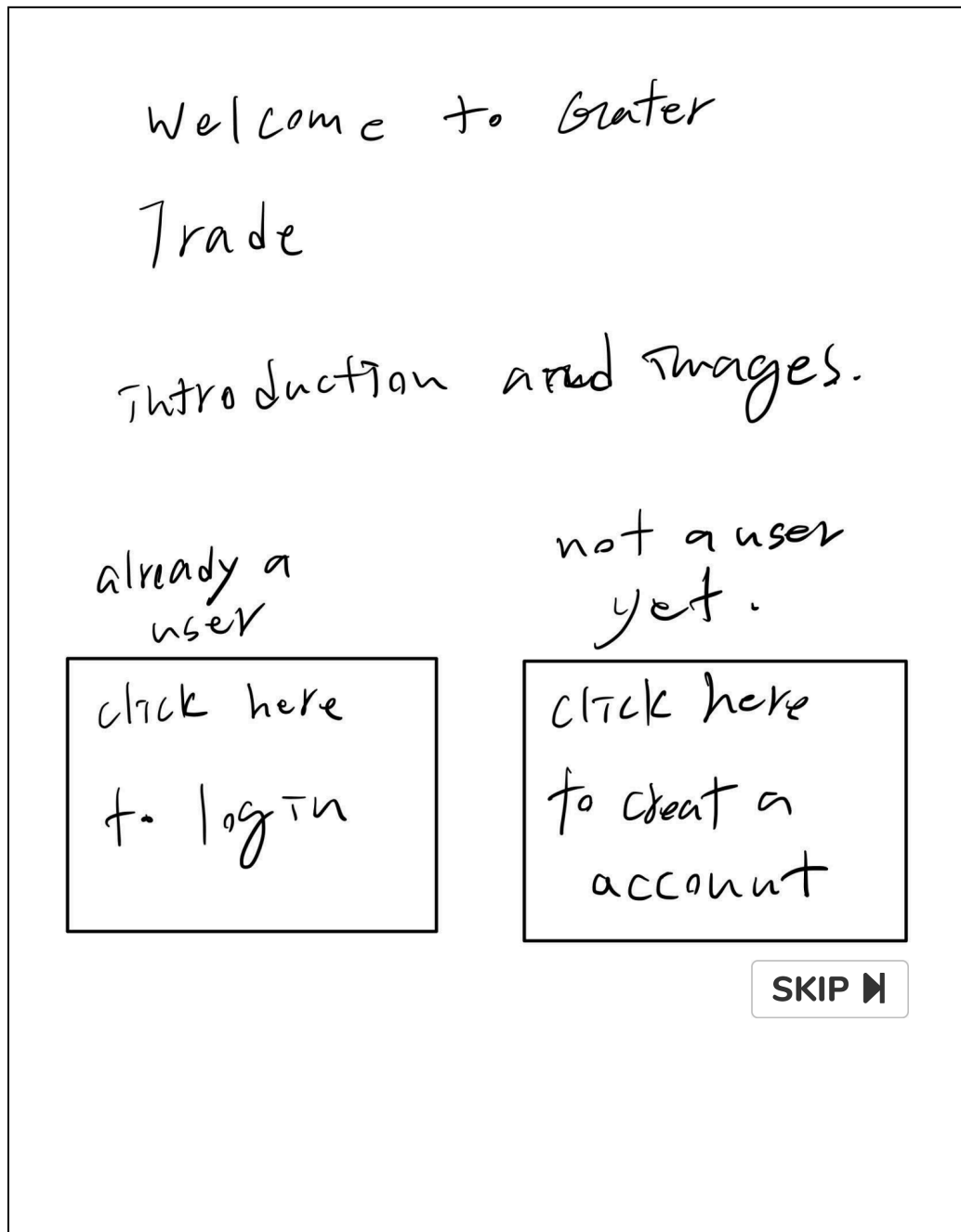
- **17. University event promotions** – The platform shall allow student clubs and organizations to post event-related listings (e.g., graduation furniture sales).

IV. UI Storyboards

Storyboard 1 - Use Case A: User Registration and Login

Emma registering with her SFSU email and logging in to access the website.

Screen 1: Home Page (Unregistered)



Emma goes onto the website and is presented with three options: logging in, registering, or skipping to view the site's listings. She decides to create an account.

Screen 2: Registration Page

(signup page)

Logo

Create an account

fill out information

First Name Last Name

Username

Email

School email only

password

creat account

click create account and go back to login page.

Emma is presented with fields for her name, username, SFSU email, and password. There is also a "Create Account" button and a note saying "School emails only" (e.g., @mail.sfsu.edu).

Screen 3: Login Page

(Login page)

Logo Login help?

Log in

user school email:

(school email only)

password:

Log IN

not user yet? | sign up

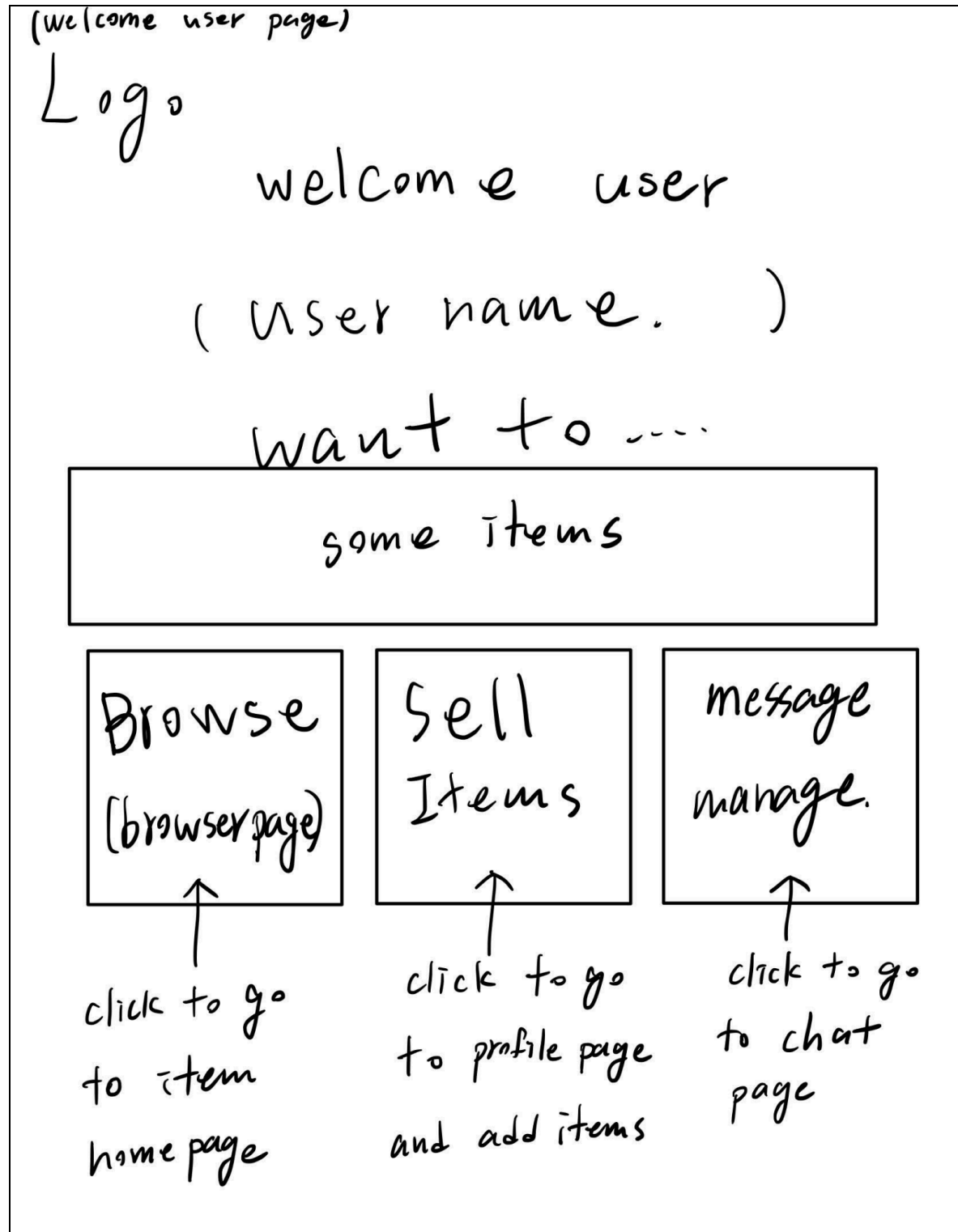
about

click Log in button
to go to welcome
user page.

click here to sign up
and turn to registration
page

Emma's account creation went through, allowing for the page to automatically redirect to the log-in page where she inserted her username and password to log in.

Screen 4: Home Page (Logged-in)



Upon successful login, Emma lands on the dashboard with key options such as "Browse Items," "Sell Now," and "Manage Messages". A featured item is at the top.

Storyboard 2 - Use Case B: Listing An Item

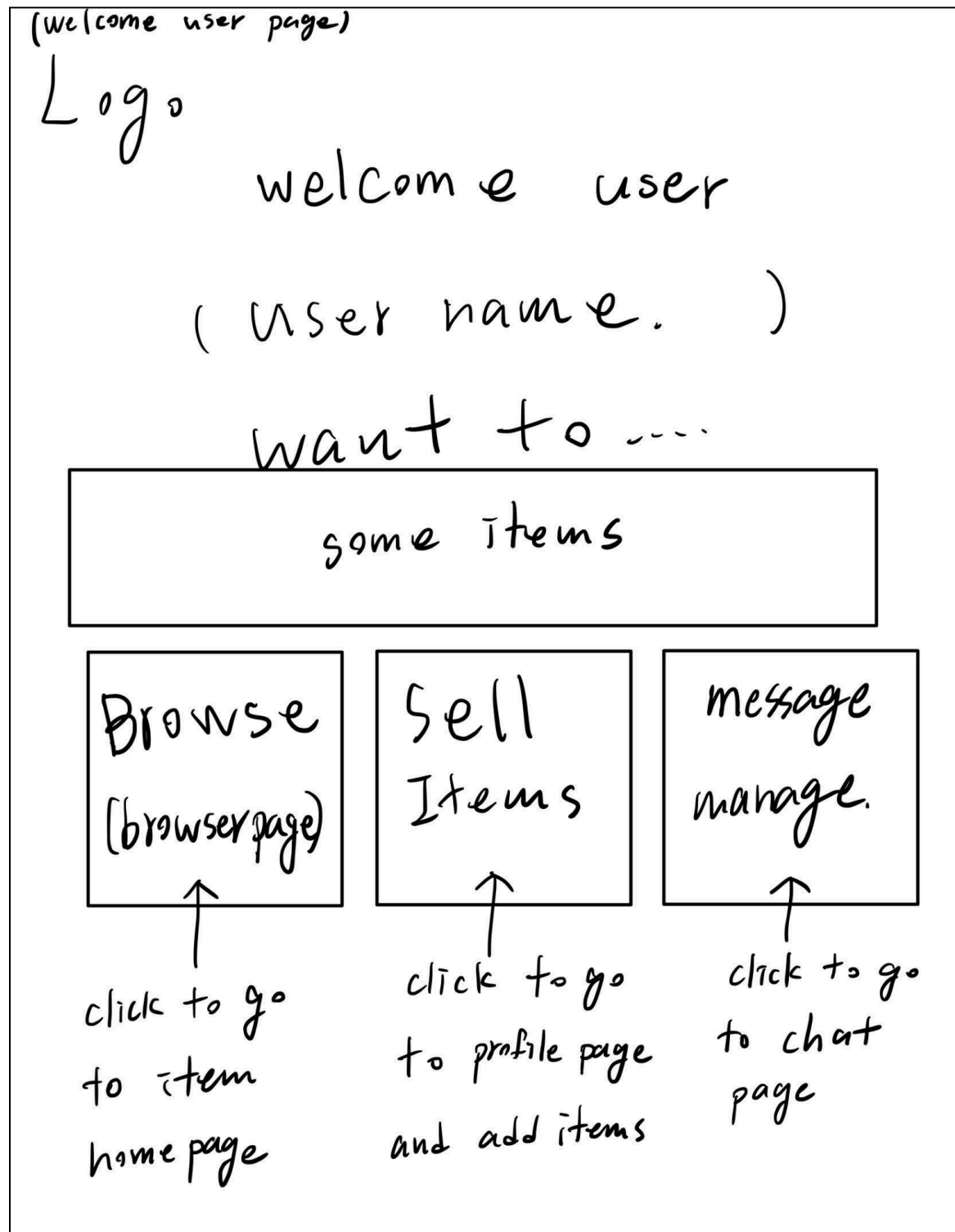
James is listing his gaming chair for sale.

Screen Page 1: Home Page (Unregistered)



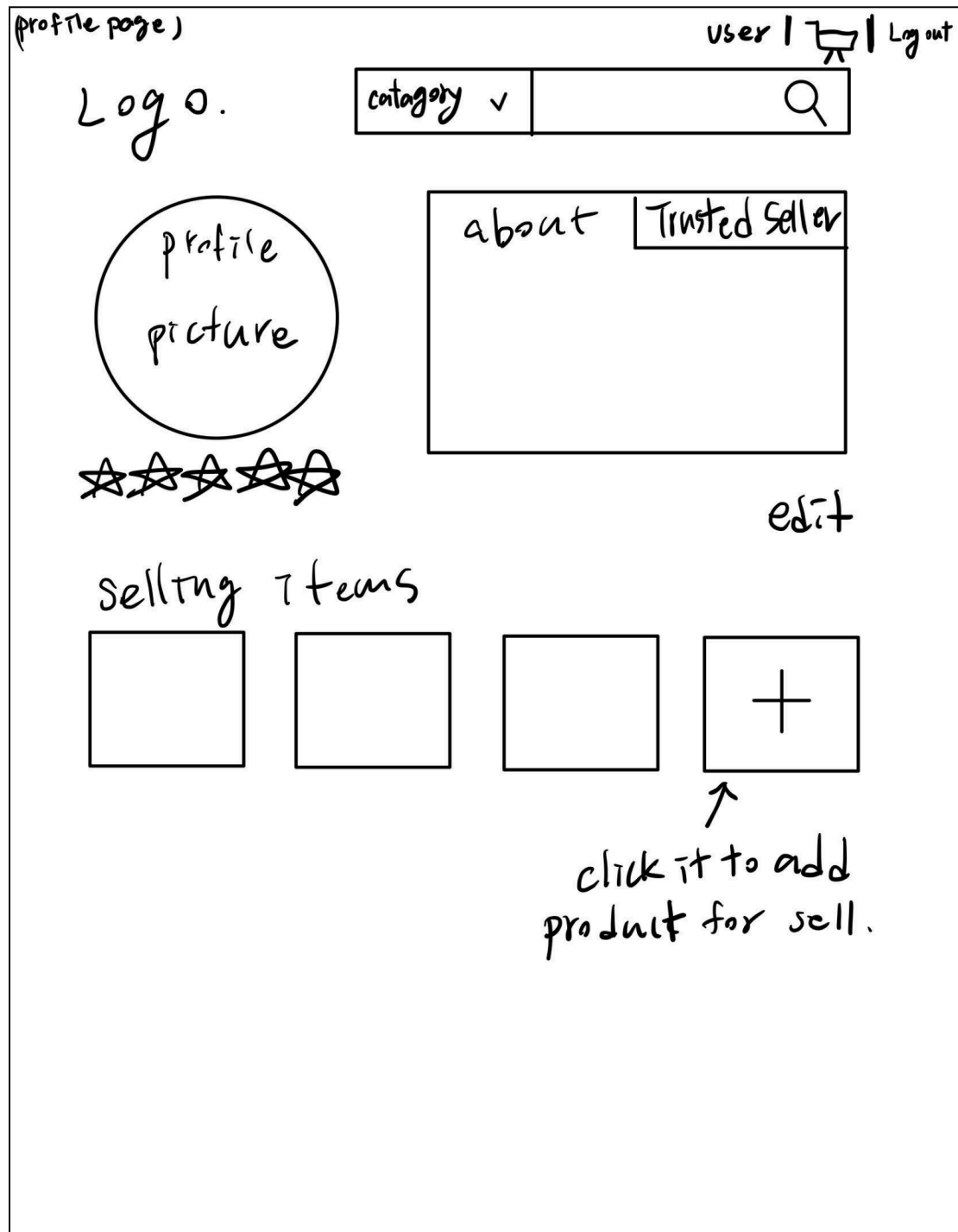
After a break from using the site, James decides to sell his gaming chair. Since his session had expired, he logs in with his existing account.

Screen Page 2: Home Page (Registered)



After logging in, James clicks on "Sell Items," which redirects him to his profile page where all his listings are displayed.

Screen Page 3: Profile Page



On his profile page, James sees his account details and current listings. He then clicks the "+" button to create a new listing for his gaming chair.

Screen Page 4: Listing Creation Page

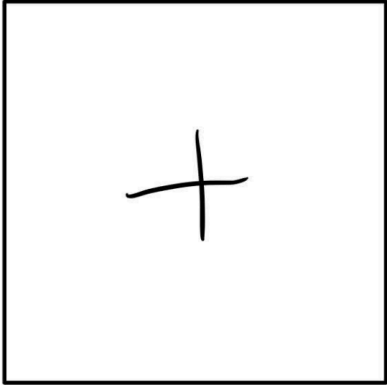
(item selling page)

user |  | Logout

Logo.

category v 

add information of the product



Item Name

color

category v

product condition

Price

available location



text

click publish
to send product
to admin
(go back to
profile page)

publish

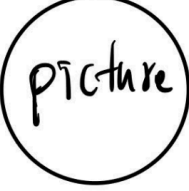
On the listing creation page, James enters the details of his gaming chair, adding a title, description, price, category, and photos before selecting a nearby available location.

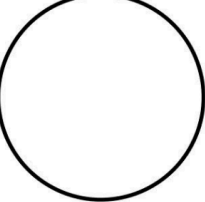
Screen Page 5: Message Hub Page

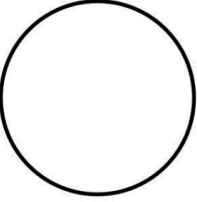
(chat contact page)

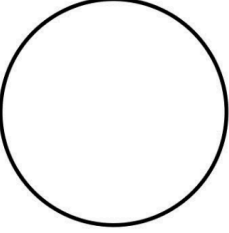
user |  | Logout

Logo.

 system message
After admin review, your listing has been published! Nowby SFSU students will now see your item. unread

 buyer 1
offer a new deal unread

 buyer 2
read

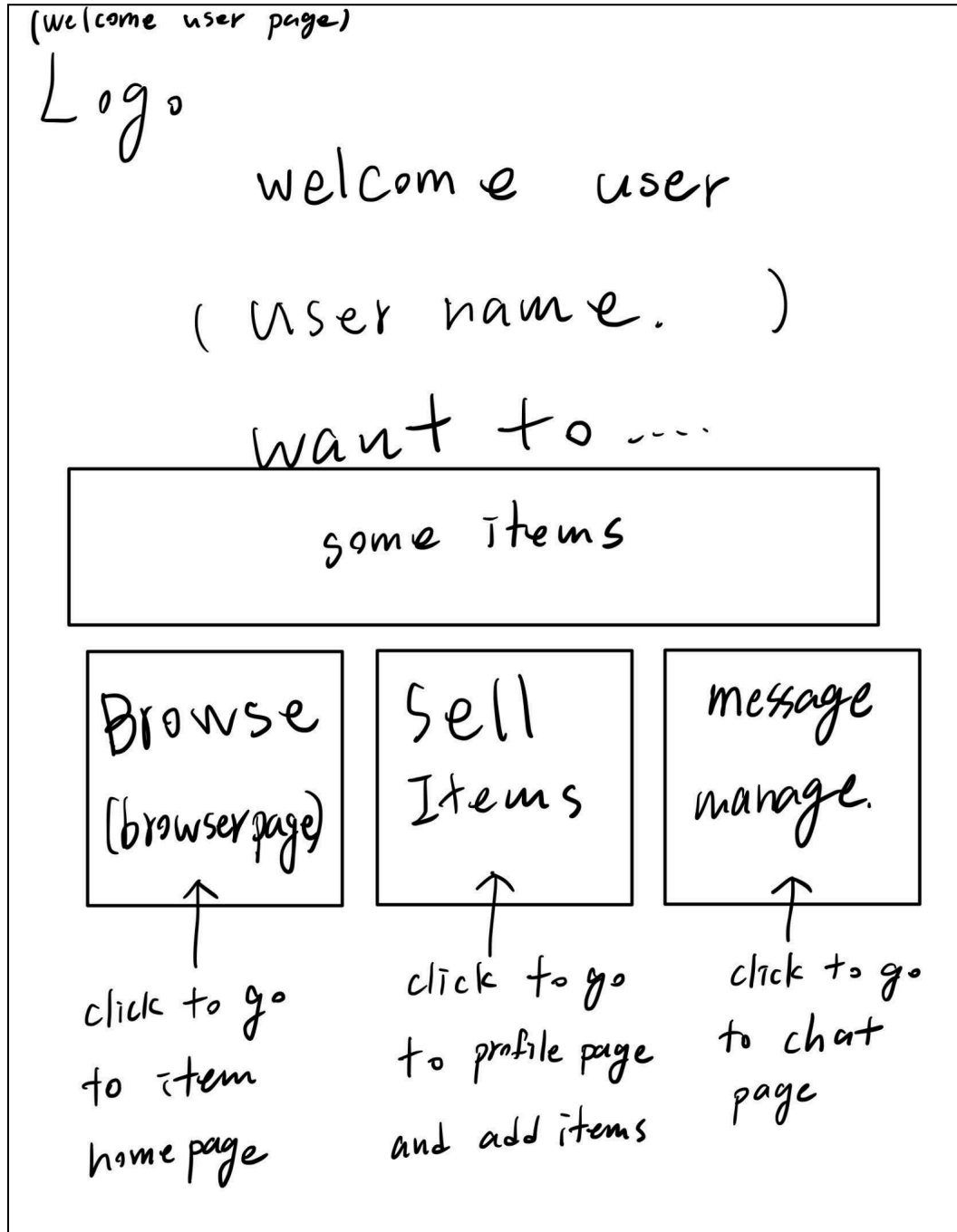
 buyer 3

After publishing the listing, James waits for an admin to review it. Once approved, he receives an automatic message confirming that his item is now live on the site and visible to potential buyers.

Storyboard 3 - Use Case C: Browsing For Items

Emma uses the site to search for a used bike and messages a seller to learn more.

Screen 1: Home Page



With Emma being a daily user, going onto the site automatically redirected her to the Home page and logged in through sessions. From here, she clicks Browse items.

Screen page 2: Browsing Page

(item home page)

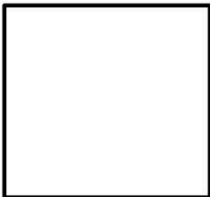
user |  | Logout

Logo. category v

categories

 Sell Item!

Shop Now

 Check Message

After being redirected to the browsing page, she used the search bar to type in “bikes” and then refined the results by selecting relevant categories.

Screen 3: Listing/Results Page

search result page)

user |  | Logout

Logo.

category v

categories	picture	info
—		
—		
—		

On the listings page, Emma browses several options and clicks on one that catches her eye to view more details.

Screen Page 4: Item Page

(item page)

user |  | Log out

Logo.

category v

Item Name

price.

select type v (if needed)

Buy Now

chat with seller

Similar Items

Customers also buy

On the item page, Emma views detailed information and sees similar listings. To make a purchase, she clicks "Buy Now," which opens a chat with the seller.

Screen Page 5: Messaging Page

user |  | Log out

Logo.

category v 

item user try
to chat with
seller info

text

seller
pic

text

item user try
to chat with
seller info

New Price

accept deal | off new deal

offer
new deal
↓
meet up
location
↙

+ 

text



Emma messages the seller. Here, they can negotiate the deal and agree on a meeting spot near campus. If they agree on a deal, both can click "Accept" to proceed to the confirmation page.

Screen Page 6: Confirmation Page

click accept or buy now/

user |  | Log out

Logo.

category v



buying location select

List of locations on campus

v

else :

type in location



warning.

public area . . .

confirm

↑

send notification
to chat for the deal
info to both user.
(place, time, price)

The seller selects a meeting spot—limited to locations near campus, and once the buyer clicks "Confirm," the system sends a message with the item details, including price, location, and time.

V. High-Level Architecture

A. Database Organization

1. Users

- id (Primary Key)
- first_name
- last_name
- email
- password_hash
- role
- created_at

2. Listings

- id
- user_id (Foreign Key to users.id)
- title
- description
- price
- category
- Image_url
- thumbnails
- status
- created_at

3. Messages

- id
- sender_id (Foreign Key to users.id)
- receiver_id (Foreign Key to users.id)
- listing_id
- content
- timestamp

4. Admin_Review_Logs

- id
- admin_id (Foreign Key to users.id)
- listing_id (Foreign Key to listings.id)
- decision (ENUM: 'approved', 'rejected', 'flagged')
- moderator_notes
- reviewed_at

5. Review

- review_id
- reviewer_id
- reviewee_id
- listing_id

- rating
 - comment
 - created_at
6. Favorite
- favorite_id
 - user_id
 - listing_id
 - timestamp
7. Reports
- report_id
 - reporter_id
 - listing_id
 - report_type
 - report_description
 - timestamp
 - review_status
8. System_Announcement
- announcement_id
 - admin_id
 - title
 - content
 - timestamp
 - status
9. Lost_and_Found
- post_id
 - poster_id
 - title
 - description
 - category
 - location_description
 - contact_method
 - image_url
 - timestamp
10. Event_Promotion
- event_id
 - poster_id
 - title
 - description
 - event_date
 - category

- image_url
- timestamp

B. Media Storage Strategy

1. Image storage will be handled via file system or AWS S3
2. Only the URL/path to the image is stored in the image_url field in the Listings table.
3. The team is not using BLOBs for media, keeping the database lightweight and performant.
4. Currently, we do not plan to store or handle video data due to limitations with free AWS

C. Search & Filter Architecture

1. We will implement search functionality using SQL Queries with the LIKE operator

- Searchable Fields:
 - (1) title
 - (2) description
 - (3) Category
- Search Logic

```
SELECT * FROM listings
WHERE title LIKE '%keyword%'
      OR description LIKE '%keyword%'
      OR category LIKE '%keyword%';
```

- Filtering and Sorting
 - (1) Handled via API query parameters

```
GET /listings?category=books&min_price=10&max_price=50&sort=newest
```

- (2) Filtering is done using WHERE clauses, and sorting via ORDER BY in the SQL query

D. Special Logic or Non-Trivial Algorithms

1. We will be implementing ranking and rating algorithms.

VI. Key Risks

Skill Risk:

Challenges relating to skills are one of the primary risks for the GatorTrade project. Certain team members may lack full-stack deployment skills, which include front-end deployment, database administration, and backend hosting. Additionally, people who are not familiar with these technologies may find it challenging to set up secure authentication, such as OAuth and SFSU email verification. We will solve this through cooperation between the team. More experienced group members will help the less experienced team members, and backend developers will investigate authentication alternatives and prototype implementations beforehand.

Teamwork risk:

Teamwork risks include potential communication breakdowns between front-end and backend teams, which could lead to integration issues, as well as unclear task distribution, which could result in redundant work or missing features. To avoid these issues, we will keep extensive API documentation using tools like Swagger or Postman and hold weekly team sync meetings via Slack or Discord. Team members will be given specific tasks and responsibilities to guarantee accountability and efficiency.

Technical risk:

Technical risks are database performance and security vulnerabilities. We can mitigate these risks by first indexing and caching correctly to ensure the utmost speed in the database. To prevent SQL injection, we will carefully add input validation and careful parameters to prevent foreign database changes.

Legal Risk:

Some Legal risks associated with managing user data, such as emails and messages, which need strict privacy protection, and with adhering to SFSU standards. We will examine SFSU's policies on student-based marketplaces and put strong encryption.

VII. Project Management

I plan to approach milestone 2 the same way I've handled all our previous assignments: by breaking the project into smaller, manageable tasks and assigning each team member a clear plan with a due date. Since we're all seniors, I have confidence in my teammates to meet their deadlines, and so far, they've consistently delivered.

For milestone 2 specifically, the back-end and front-end teams will each focus on their respective responsibilities, which will eventually integrate as we progress toward project

completion. The back-end team will handle the database and develop the JavaScript search function, while the front-end team will focus on designing and building the web pages.

As the team lead, I act as the communication bridge between both teams, ensuring that everything runs smoothly and that we stay on schedule. In every meeting, I assign tasks, document them in our Discord server, and provide short but reasonable deadlines. This gives everyone a clear reference point if they need to revisit their responsibilities.

Once all the tasks are completed, either I or another teammate will thoroughly review and edit the document to correct any formatting or grammatical issues. Once we're confident in the final product, we submit it.

VIII. Use of GenAI

Brainstorming Ideas:

ChatGPT was used extensively during the early planning phase to brainstorm potential project risks, feature ideas, and UI/UX flow suggestions. It helped propose several use-case scenarios and alternative flows that shaped the initial direction of the app. Additionally, it offered input for early-stage storyboard suggestions, such as how users would navigate between listing pages, messaging, and admin moderation.

Coding:

ChatGPT provided support in debugging and troubleshooting frontend and backend code. While it currently isn't used as much right now for milestone 2 part 1, we predict the use of it will go up as we program more in part 2.

Writing and structure:

ChatGPT was used to improve the clarity, grammar, and professional tone of milestone documents, presentation slides, and report content. It assisted in rewording long or unclear sentences, ensuring smoother flow between sections, and standardizing terminology across documents.

List of Main Data Items & Functional Requirements:

ChatGPT assisted in organizing and naming core data entities such as *User*, *Listing*, *Message*, and *Review*. It helped in defining the structure of each data item, including value examples and required/optional status.

Rating (out of 5):

- **Brainstorming Ideas** – 2 (Low)
- **Code Assistance** – 2 (Low)
- **Main data items and functional requirements** – 4 (Medium)
- **Writing & Formatting Help** – 5 (High)

Additional Information:

- ChatGPT Used: 3.5/4.0
- Pretty good to expand upon ideas already written down.

Example Prompts:

- “Can you format this so it follows what the question is asking for?”
- “Simplify the formatting.”
- “Should I add more things to priorities 2 and 3?”

IX. Team Lead Checklist

- a. So far, all team members are fully engaged and attending team sessions when required - DONE
 - b. Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing - ON TRACK
 - c. Team reviewed suggested resources before drafting Milestone 2 - DONE
 - d. Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission - DONE
 - e. Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission - DONE
 - f. Team shared and discussed experience with GenAI tools among themselves - DONE
- **END OF MILESTONE 2 DOCUMENT**

SW Engineering CSC 648-848 Spring 2025

Milestone 3:

Team Number: 08

Meeting Date: 4/23/2025

Version:	Date Submitted:	Date Revised:
M3	4-26-2025	N/A

Summary of feedback:

UI:

Overall, the team is on track to complete the assignment on time. Most UI feedback involves minor tweaks rather than major redesigns. A few big features still need to be added, such as a filtering system, a more dynamic map, and more efficient image loading with bandwidth in mind. Beyond that, the remaining updates are small and straightforward, like adding required fields in registration forms, bolding important words in the contacts section, updating the navigation bar for login and registration, and making minor adjustments to the homepage. Since these revisions are easy to implement, it's safe to say we're ready to start integrating the frontend and backend.

Code and architecture:

N/A

GitHub Usage:

N/A

Database:

N/A

Teamwork and Risk Management:

Overall, teamwork is good, the professor mentioned that the team is "on track" so there isn't much to be worried about. Our biggest risk right now is just running out of time, but so far, everything is going well. I believe that as team lead, we will only start changing things up if we mess up badly.

- Architecture Review -

P1 Features:

Committed Functions

- Browsing Items: Anyone can browse all items, including categories, prices, images, and descriptions.
- Searching & Filtering: Search by keyword, filter by category, and set a price range to narrow down results.
- Viewing Item Details: View full item info like images, description, price, and posting time.
- Student Registration: Only users with a valid SFSU email can register.
- Login Access: Registered users can securely log in to access full features.
- Posting Items for Sale: Users can create listings with images and details (admin-reviewed before posting).
- Managing Listings: Users can delete their listings or mark them as sold.
- In-Site Messaging: Buyers and sellers communicate through a private chat system.

Additional Comments;

- None

- END OF MILESTONE 3 DOCUMENT

SW Engineering CSC 648-848 Spring 2025

Project Name: GatorTrade

Team 08

Team Lead: Jonathan Carbajal - Jcarbajal@sfsu.edu

Backend Lead: Christian Kurt Balais - cbalais@sfsu.edu

Frontend Lead: Tung-Ying Lee - nlee@sfsu.edu

Frontend Lead 2: Chia-Tung Liu - cliu26@sfsu.edu

Github Master: Luis Moreno - lmoreno10@sfsu.edu

Milestone 4

Date: 05/20/2025

History Table:

Version:	Date Submitted:	Date Revised:
M4	5/20/2025	5/20/2025

I. Product Summary:

GatorTrade is a safe, student-focused online marketplace. Understanding the drawbacks and dangers of popular websites like Facebook Marketplace, Craigslist, OfferUp, and eBay, our team developed GatorTrade to specifically address the difficulties SFSU students encounter when purchasing and selling daily necessities like furniture, electronics, and textbooks. GatorTrade's closed-network environment, which only allows individuals with validated SFSU email addresses to join, is what makes it unique. This considerably lowers the possibility of scams, spam, or anonymous malicious users by guaranteeing that all interactions occur inside a recognized and trusted academic community. GatorTrade was created with student safety and convenience as its top priorities, in contrast to larger platforms that place more emphasis on reach and ad revenue.

Major Committed Functions: P1

- **Browsing Items:**
Unregistered users can browse all items, including categories, prices, images, and descriptions.
- **Searching & Filtering:**
Users can search by keyword, filter by category, and set a price range to narrow down listings.
- **Viewing Item Details:**
Everyone can view full item information — including images, full descriptions, prices, and posting time.
- **Student Registration:**
New users can sign up using their valid university (SFSU) email address.
- **Login Access:**
Registered users can log in to access posting and messaging features.
- **Posting Items for Sale:**
Logged-in users can list items by submitting details and images, which go through admin approval before being published.
- **Managing Listings:**
Sellers can delete their listings or mark items as sold when no longer available.
- **In-Site Messaging:**
Buyers can contact sellers through a private in-platform messaging system.
- **Seller Inbox & Notifications:**
Sellers are notified via automated email (sent to their SFSU inbox) when a buyer sends them a message for the first time.
- **Profile Management:**
Users can update some of their personal information

- **Safe Meetup Locations:**

The platform suggests secure on-campus meetup spots that users can choose for safe exchanges in messages. Choosing a location sends a prewritten message.

URL: <http://18.117.162.109/>

II. Usability Test Plan:

A. Test Objectives

- Evaluate how effectively and efficiently users can search for listings based on category and text input. The goal is to assess the intuitiveness of the UI and how well users understand and complete the task.

B. Test Background and Setup

- System setup: Deployed React + Express app, backend connected to MySQL DB.
- Tester Hardware: Desktop/laptop with a modern browser (Chrome, Firefox).
- Target Users: SFSU students interested in peer-to-peer item exchange.
- URL: <http://<your-ec2-ip>>
- Environment: Home setting, no cameras or monitoring; no training required.

C. Usability Task Description

- You are a student looking to buy a used textbook. Please:
 - a) Go to the website.
 - b) Use the category dropdown to select "Books".
 - c) In the search bar, type a relevant keyword (e.g., "Calculus").
 - d) Click the "Search" button.
 - e) View the results, and try to find the most relevant listing.
 - f) Tell us how confident you are that you found what you sought.

D. Evaluation of Effectiveness

- Effectiveness will be measured by:
 - a) Whether the correct results are returned based on the input
 - b) Whether users were able to complete all tasks
 - c) Observations of any mistakes, confusion, or abandonment

E. Evaluation of Efficiency

- Efficiency will be measured by:
 - a) Time taken from task start to successful completion
 - b) Number of clicks and interactions
 - c) User hesitation or repeated actions

F. Evaluation of User Satisfaction

- Testers will answer these after the task:
 - a) "It was easy to find and use the search function."
→ 1 (Strongly Disagree) to 5 (Strongly Agree)

- b) "The search results were relevant to my input."
→ 1 to 5 scale
- c) "I would use this site to search for items again."
→ 1 to 5 scale

G. GenAI Use:

- Tool: ChatGPT (GPT-4)
- Use: Guide on how to do the test plan, refined Likert questions
- Prompt Example: "What do I need to draft a usability test plan?"
- Utility: High

III. QA Test Plan and Testing:

1. Test Objectives

To verify that the search functionality:

- Accurately returns results based on entered keywords
- Displays correct and relevant school entries
- Handles incorrect or empty inputs gracefully
- Functions properly across major web browsers

2. HW and SW Setup

- **Hardware:** Standard laptop (Intel i9 ,16GB RAM)
- **Operating Systems:** Windows 11
- **Browsers Used for Testing:**
 - Google Chrome v123
 - Safari
- **URL:** <http://localhost:5000/> , <http://18.117.162.109/>

3. Feature to Be Tested

Search Function on Homepage and Search Results Page

The user enters a school name or keyword into the search bar. The system returns matching results based on school name, location, or tags.

4. QA Test Plan Table

Test #	Test Title	Test Description	Test Input	Expected Output	Chrome	Safari
1	Basic Search	Tests keyword match the result	chair	Results showing all chairs from the market	PASS	PASS
2	Invalid/Empty Search	Tests handling of blank or unrelated inputs	asdfgh / blank	Message: “No results found”, for blank it just show all the result on the website	PASS	PASS
3	Case-insensitive Search	Check if uppercase/lowercase query returns same results	BOOK	Same results as book	PASS	PASS
4	Partial Match	Input a partial word to test for autocomplete or partial string search match	ch	Results include items where either the name or description contains “ch”.	PASS	PASS
5	Search Result Click-through	Click on a search result and verify redirection to correct items	Click selling item	Redirects to full item info page	PASS	PASS
6	Search Input Limit	Ensure the input does not accept more than 40 characters	A string longer than 40 characters	An alert shown to user.	PASS	PASS

5. Test Results Summary

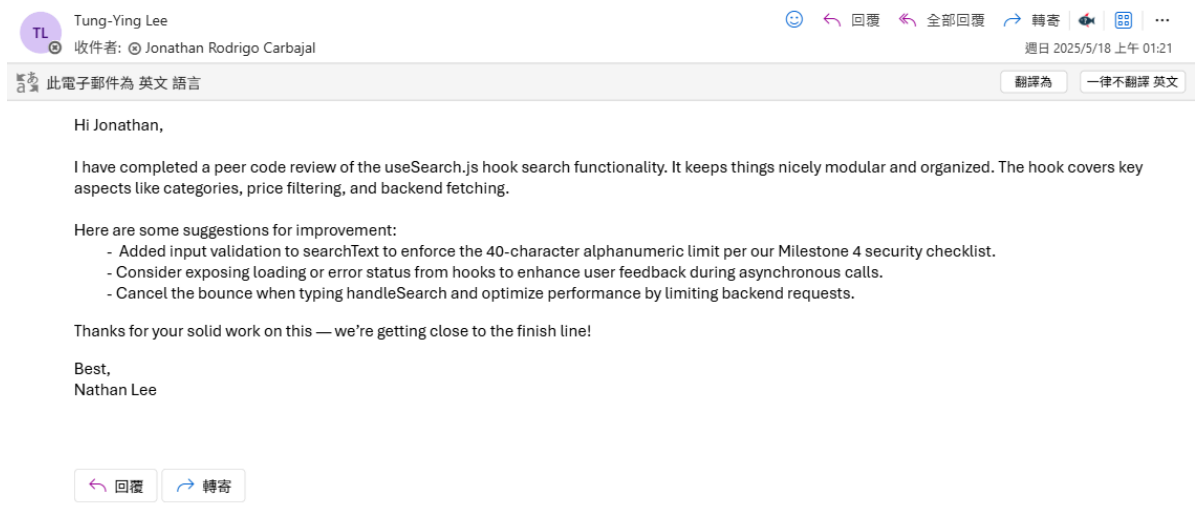
- Testing was executed manually on both Chrome and Firefox.
- All five test cases passed successfully on both browsers.

- No functional bugs or UI rendering issues were observed.

6. GenAI Use Summary

- **Tool Used:** ChatGPT (GPT-4)
- **Usage:** Assisted in drafting the test plan structure, test case ideas, and refining test input/output expectations.
- **Prompts Used:**
 - “Help write a QA test plan table for a school search website feature.”
 - “What are good test cases for a search input form?”
- **Benefit:** Helped ensure comprehensive test coverage, clearly written cases, and professionally formatted table.
- **Utility Ranking: HIGH** – Saved time and improved clarity of documentation.

IV. Peer Code Review:



```

1  import { useState } from 'react';
2
3  export function useSearch(initialCategory = '', initialText = '', initialResults = []) {
4    const [searchCategory, setSearchCategory] = useState(initialCategory);
5    const [searchText, setSearchText] = useState(initialText);
6    const [results, setResults] = useState(initialResults);
7    const [minPrice, setMinPrice] = useState(0);
8    const [maxPrice, setMaxPrice] = useState(1000);
9
10   const handleSearch = async (
11     categoryParam = searchCategory,
12     textParam = searchText,
13     min = minPrice,
14     max = maxPrice
15   ) => {
16     try {
17       const params = new URLSearchParams({
18         category: categoryParam,
19         query: textParam,
20         minPrice: min,
21         maxPrice: max,
22       });
23
24       const response = await fetch(`/items/search?${params.toString()}`);
25       const data = await response.json();
26
27       if (response.ok) {
28         setResults(data);
29       } else {
30         console.error("Search error:", data.error);
31       }
32     } catch (error) {
33       console.error("Fetch error:", error);
34     }
35   };
36
37   return {
38     searchCategory,
39     setSearchCategory,
40     searchText,
41     setSearchText,
42     results,
43     setResults,
44     minPrice,
45     setMinPrice,
46     maxPrice,
47     setMaxPrice,
48     handleSearch,
49   };
50 }

```

File Reviewed: src/hooks/useSearch.js

Functionality: Search hook for managing query input, category filter, price range, and backend fetch

What's Done Well

- Clarity and readability: The code is well-structured and clean. You used consistent formatting and spacing.
- Naming conventions: Function and variable names (such as handleSearch, setSearchText, and minPrice) are self-explanatory, following the camelCase convention.
- Functional cohesion: State management is tightly grouped, making it easy to extend filtering or updating logic.
- Modularity: Wrap your search logic in reusable hooks to make your app scalable and DRY.

Improvement Suggestions

- Input Validation:
 - This hook does not enforce any character length restrictions on searchText. Add the following checks:
if (textParam.length > 40 || /^[^a-zA-Z0-9]/.test(textParam)) return;
- Error feedback:
 -
 - If an API fetch fails or returns an error, the user never sees it.
 - Consider passing an error message or a boolean flag back from handleSearch so your UI can display that information.
- Performance - Debounce:
 - Fast input will send multiple GET requests.
 - It is recommended to wrap handleSearch in a debounce utility when integrating into components to reduce unnecessary API calls.
- Missing loading status:
 - The user has no visual cue while the results are loading.
 - It is recommended to add isLoading to the return status of your hook and toggle it during asynchronous fetching.

GenAI Code Review Summary

Tool: ChatGPT

Prompt Used:

Hi, I'm a senior Computer Science student reviewing React custom hooks for search functionality in my team's marketplace app. Can you help me evaluate the code to see if there are improvements in structure, performance, input validation, and overall reusability? Please highlight suggestions that would be helpful for app expansion or for others to use later.

GenAI Review Results:

- Separate API Logic from State Logic
- "It would be a good idea to move the fetch call into a separate utility or service file. That way, if the API endpoint changes or is reused elsewhere, you won't have to update the hook itself."

Expose Reset Functionality

- "Add a resetFilters() function to clear search fields and filters. This would make it easier to add a reset button in the UI and improve usability."

Transform API Response

- "You could shape the response data before setting it into state. For example, map or filter fields you'll actually need in the UI. This will make the UI rendering cleaner and more maintainable."

Improve Documentation

- “Try adding brief comments for each state variable and return value, especially `handleSearch()`. Even short inline descriptions help other developers (or your future self) understand it faster.”

Scalability Note

- “As your app scales, you might want to include pagination or sorting support. Structuring the hook to handle those now (even as optional params) could save refactoring later.”

GenAI Utility RANK: HIGH

This GenAI review provided a different angle from the peer review by:

- Focusing on code reusability, modularity, and growth readiness
- Suggesting improvements that make the hook more maintainable and scalable
- Offering ideas you can use even if the current version works fine

It complements the peer review by avoiding overlap and showing how the hook could evolve as the project grows or enters production.

V. Self-Check for Security:

Asset To Be Protected	Types of possible/expected Attacks	Consequences of a Security Breach	Strategy to mitigate/protect the asset
Registration	Non-accepted values inserted.	Unauthorized accounts created.	Regex checks if the school email is used, frontend AND backend both check if the inputs are acceptable.
Full Name	Impersonation, phishing	Identity misuse	User review, deletion from database (erase unaccepted accounts).
Passwords	Login Brute-force	Unauthorized access	Passwords are hashed using bcrypt in the backend before saving to the DB.
Messages	Unauthorized access, URL manipulation	Privacy breach, access to private communications	Backend checks both sender & recipient IDs before returning messages. URL tampering fails in the backend.
Listing (Status)	Fake or malicious posts	Scams or misinformation	Admins must approve listings before public display. Status defaults to <i>pending</i> .
Listings (ownership)	Unauthorized UserID reassignment	Data spoofing, impersonation	Only DB admins can reassign userIDs.
Listing (images)	Inappropriate content, non-image uploads	Content policy violations	All images are reviewed before listing approval. The backend limits file type & size.
Session State	Session hijack, improper logout	Unauthorized actions from stale sessions	Sessions expire after 24 hours or on server restart. No persistent login allowed.
Unauthorized Deletion	User deletes someone else's listing	Loss of data, abuse of access	Backend checks listing ownership (userID) before allowing deletes. Only admins can delete listings in workbench.

Additional Nodes:

- **Passwords are encrypted** using bcrypt before being saved in the database.
- Search Bar ONLY accepts 40 alphanumeric characters max
- Registration checks if the email ends with "sfsu.edu".
- Dummy terms of service are in registration, and must be accepted.

VI. Self-Check for Original Non-functional specs:

1. The application shall be developed, tested, and deployed using tools and cloud servers approved by Class CTO and as agreed in M0. **DONE**
2. The application shall be optimized for standard desktop/laptop browsers; e.g., it must render correctly on the two latest versions of two major browsers. **DONE**
3. All or selected application functions shall render well on mobile devices (no native app to be developed). **DONE**
4. Posting of sales information and messaging to sellers shall be limited only to SFSU students. **DONE**
5. Critical data shall be stored in the database on the team's deployment server. **DONE**
6. No more than 50 concurrent users shall be accessing the application at any time. **DONE**
7. The privacy of users shall be protected. **DONE**
8. The language used shall be English (no localization needed). **DONE**
9. The application shall be easy to use and intuitive. **DONE**
10. The application shall follow established architecture patterns. **DONE**
11. The application code and its repository shall be easy to inspect and maintain. **DONE**
12. Google Analytics shall be used. **DONE**
13. No e-mail clients or chat services shall be allowed. Interested users can only message sellers via in-site messaging. One round of messaging (from user to seller) is enough for this application. **DONE**
14. Pay functionality, if any (e.g., paying for goods and services), shall not be implemented nor simulated in UI. **DONE**
15. Site security: Basic best practices shall be applied (as covered in the class) for main data items. **DONE**

16. Media formats shall be standard as used in the market today. **DONE**
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools. **DONE**
18. The application UI (WWW and mobile) shall prominently display the exact text on all pages: "SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only" at the top of the WWW page Nav bar. (It is important to not confuse this with a real application). You have to use this exact text without any editing. **DONE**

VII. Use of GenAI:

Brainstorming Ideas:

ChatGPT was used to help brainstorm and refine final milestone features, including search filtering logic, user session handling, and enhancements to the item listing and header components. It provided suggestions for category design, helped identify potential pain points in filtering implementation, and assisted in defining the logic behind secure login state handling.

Coding Support:

Gen ai was used to help with bugs and error codes in the terminal. It was used a lot to learn react and check code for errors since it was our first time with react. Helped with troubleshooting filtering. Mostly used for guidance and help

Writing and Structure:

ChatGPT was used to clarify error messages, simplify conditional logic, and improve the readability of routes and controller files. Additionally, it was instrumental in helping refine report content and writing the product summary by expanding initial draft content with clearer organization and a more polished tone.

Functional Guidance:

It helped identify a clean approach to integrating price filters without disrupting existing search logic and guided the integration of dynamic header states using session-based login detection. ChatGPT also provided clear direction on how to style user-dependent components using conditional rendering.

- **END OF MILESTONE 4 DOCUMENT**

III. Product Screenshots:

SF Gator Trade

All

Search for items...

[Create Post] Sign In

Sign In

Email

you@sfsu.edu

Password

Enter your password

Sign In

Don't have an account? [Sign up](#)

SF Gator Trade

All

Search for items...

[Create Post] Sign In

Create an Account

First Name *

Jane

Last Name *

Doe

Email * (school email only)

you@sfsu.edu

Password *

Enter a password

Confirm Password *

Re-enter password

☐ I accept the [Terms of Service](#) *

☐ I accept the [Privacy Policy](#) *

Sign Up

Already have an account? [Sign In](#)

SF Gator Trade

All

Search for items...

Create Post

Sign In

Deals

Popular Picks

Books

Electronics

Clothing

Dorm Setup

School Supplies

Welcome to Gator Trade

SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only

Sell An Item!

Featured Items



Office Chair

\$129.99



Harry Potter and the Sorcerer's Stone

\$19.99



Inception (Blu-ray)

\$14.99



Denim Jacket

\$39.99

See more

Newest Postings



Logitech G PRO X SUPERLIGHT 2 Wireless Gaming Mouse

\$99.99



Old Backpack

\$22.22



Desk and Chair

\$2.00



Apple AirPods Pro (2nd Gen)

\$149.99

See more

Early Bird Finds



Harry Potter and the Sorcerer's Stone

\$19.99



Denim Jacket

\$39.99



Wireless Headphones

\$59.99



Office Chair

\$129.99

See more

Recommend to Freshmen



Denim Jacket

\$39.99



Desk and Chair

\$2.00



Inception (Blu-ray)

\$14.99



Logitech G PRO X SUPERLIGHT 2 Wireless Gaming Mouse

\$99.99

See more

© 2025 Gator Trade - Group 8

60

Results:

Sort By

Max Price

\$1000

Apply Filters

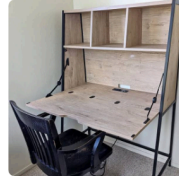
12 results found



Logitech G PRO X
SUPERLIGHT 2 Wireless
Gaming Mouse
\$99.99



Old Backpack
\$22.22



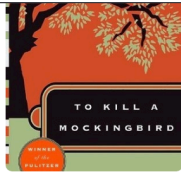
Desk and Chair
\$2.00



Apple AirPods Pro (2nd
Gen)
\$149.99



To Kill a Mockingbird (1962
Film)
\$9.99



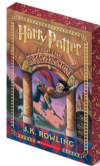
To Kill a Mockingbird
\$12.50



Inception (Blu-ray)
\$14.99



Acoustic Guitar
\$89.99



Harry Potter and the
Sorcerer's Stone
\$19.99



Office Chair
\$129.99



Wireless Headphones
\$59.99



Denim Jacket
\$39.99



Chia-Tung Liu

★ Unrated

[Log Out](#)

About

Trusted Seller

Chia-tung tung tung sahur

Email: cliu26@sfsu.edu

Member since: 2025年5月

Total Items Listed: 1

Total Items Sold: 0

[Edit](#)

Selling Items

Logitech G PRO X SUPERLIGHT 2 Wireless
Gaming Mouse
\$99.99
Approved

[Delete](#)[Mark as Sold](#)

Selling Items

Logitech G PRO X SUPERLIGHT 2 Wireless
Gaming Mouse
\$99.99
Approved

[Delete](#)[Mark as Sold](#)

Messages



Jonathan C

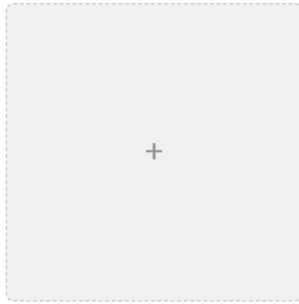
[Delete](#)

For Apple AirPods Pro (2nd Gen)

Would you like to meet at [SFSU Cafe Rosso](https://www.google.com/maps/search/?api=1&query=SFSU%20C)



Upload Selling Item



Upload 1 image

Item Name *

Category *

Preferred Location *

Price (\$) *

Product Condition *

Description *

[Publish](#)

Chat with Jon C

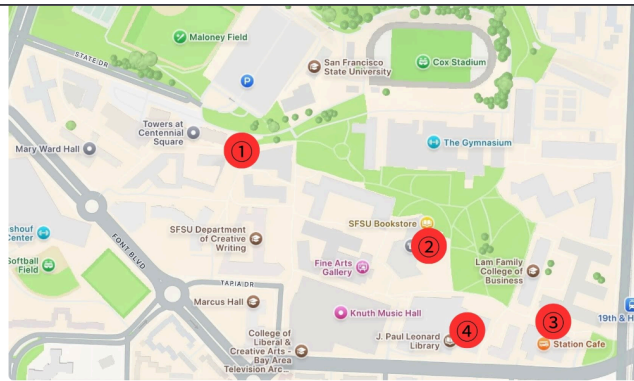
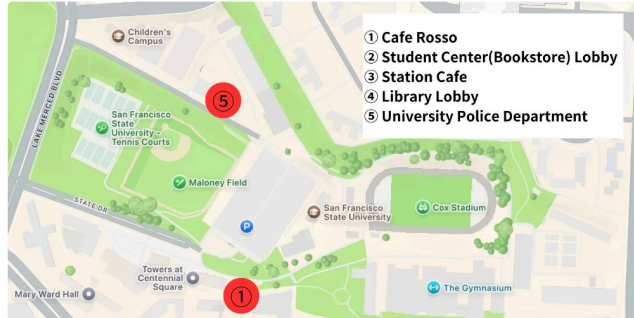
No messages yet.

[Set Up Location](#)[Send](#)

Buying Location Select

List of Locations on Campus

-- Choose a location --



⚠ Warning: Please meet in a public, safe area on campus.

Confirm

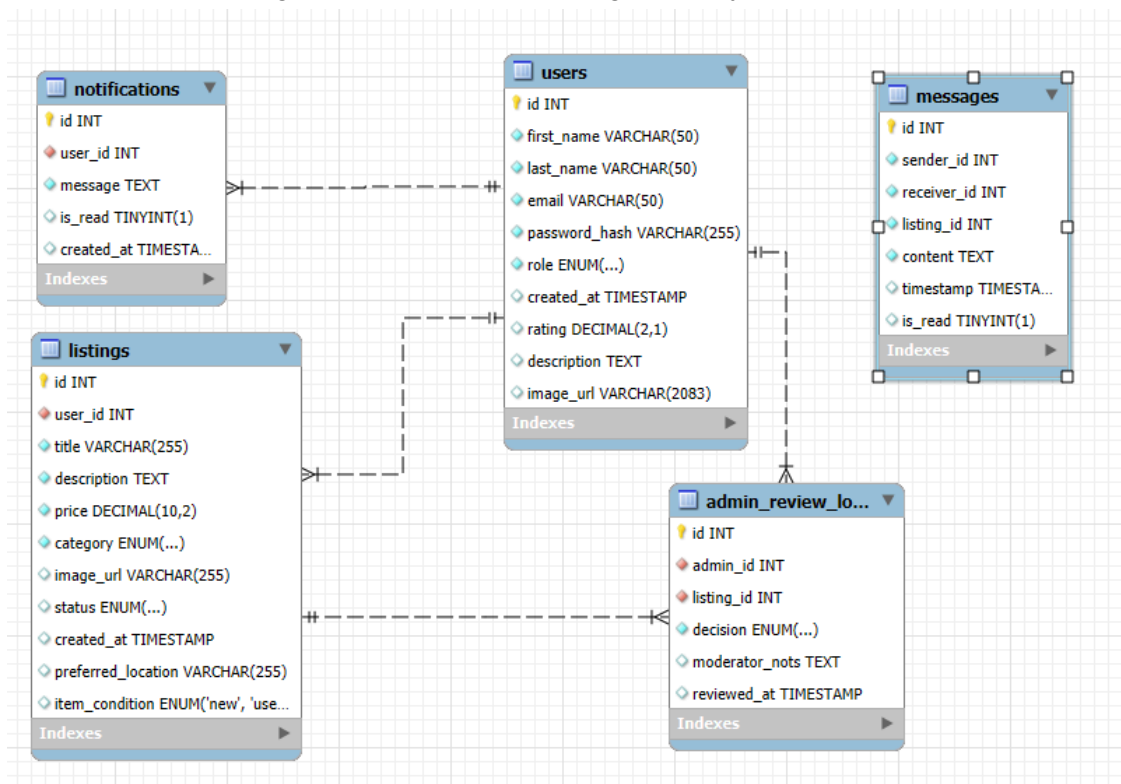
IV. Database Organization:

Result Grid		Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
	id	sender_id	receiver_id	listing_id	content	timestamp	is_read		
▶	1	6	7	19	Hey is the Leaf still for sale?	2025-05-10 05:50:41	0		
	2	6	7	19	Would you like to meet at Student Center Lobby?	2025-05-10 05:53:33	0		
	15	9	7	19	Hi, is this still available	2025-05-12 04:52:12	0		
	16	9	7	19	Do you receive my letter?	2025-05-12 04:54:31	0		
	17	9	7	19	Hola my friend	2025-05-12 05:05:32	0		
	18	9	7	19	hi	2025-05-12 05:06:59	0		
	19	7	6	19	yes	2025-05-12 05:08:20	0		
	20	7	6	19	Would you like to meet at Cafe Rosso?	2025-05-12 05:08:28	0		
	21	7	6	19	Would you like to meet at University Police Dep...	2025-05-12 05:11:48	0		
	44	14	1	9	burppp	2025-05-13 07:25:46	0		
	76	8	1	10	Would you like to meet at [SFSU Station Cafe](...	2025-05-15 02:16:47	0		
	78	8	1	10	Would you like to meet at [SFSU University Polic...	2025-05-15 02:17:39	0		
	79	8	1	10	Would you like to meet at [SFSU Cafe Rosso](ht...	2025-05-15 04:49:04	0		
	82	5	1	8	hi there	2025-05-17 07:39:13	0		
	93	7	5	32	Would you like to meet at [SFSU University Polic...	2025-05-17 22:29:42	0		
	94	5	7	32	Hell nah	2025-05-17 22:29:50	0		
	95	5	7	32	Would you like to meet at [SFSU Station Cafe](...	2025-05-17 22:30:16	0		
	96	7	5	32	Would you like to meet at [SFSU Station Cafe](...	2025-05-17 22:30:18	0		
	98	5	7	34	hey is this item still for sale	2025-05-19 02:07:58	0		
	99	7	5	34	Would you like to meet at [SFSU Library Lobby](...	2025-05-19 02:10:18	0		
	100	11	16	43	Is this available?	2025-05-20 01:33:29	0		
	101	16	11	43	no	2025-05-20 01:33:38	0		
	102	16	11	43	Would you like to meet at [SFSU Cafe Rosso](ht...	2025-05-20 01:33:54	0		
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL		

Result Grid		Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:		
	id	first_name	last_name	email	password_hash	role	created_at	rating	description	image_url
▶	1	Jonathan	C	test@sfsu.edu	password	user	2025-04-02 23:25:11	NULL	NULL	NULL
	2	J	C	jon@sfsu.edu	\$2b\$10\$51sKyv...	user	2025-05-05 00:20:12	NULL	NULL	NULL
	3	Test 3	D	sf@sfsu.edu	\$2b\$10\$nYWJft...	user	2025-05-05 00:26:19	NULL	NULL	NULL
	4	d	d	email@sfsu.edu	\$2b\$10\$TRqOu...	user	2025-05-05 00:27:16	NULL	NULL	NULL
	5	Jon	C	jon123@sfsu.edu	\$2b\$10\$nfCGp...	user	2025-05-05 00:34:39	4.5	Jon's alt account	user_5_1...
	6	Alexis	P	alexis123@sfsu....	\$2b\$10\$7TZq9...	user	2025-05-09 18:35:49	NULL	lalala	NULL
	7	Jonathan	Carbajal	jcarbajal@sfsu.edu	\$2b\$10\$QExlN...	user	2025-05-09 21:49:13	NULL	ye	user_7_1...
	8	Chia-Tung	Liu	cliu26@sfsu.edu	\$2b\$10\$7MffAT...	user	2025-05-09 22:10:07	NULL	Chia-tung tun...	user_8_1...
	9	Nathan	Lee	nlee@sfsu.edu	\$2b\$10\$Pr3t2...	user	2025-05-09 22:56:42	NULL	NULL	user_9_1...
	10	Reg	Example	12345@sfsu.edu	\$2b\$10\$9RLaD...	user	2025-05-11 22:42:10	NULL	NULL	NULL
	11	luis	m	luis@sfsu.edu	\$2b\$10\$AVPrBX...	user	2025-05-12 00:33:07	NULL	NULL	NULL
	14	PFP CHECK	YE	test5@sfsu.edu	\$2b\$10\$XvOCB...	user	2025-05-13 00:17:38	NULL	test	user_14_...
	15	Jake	The Dog	jake@sfsu.edu	\$2b\$10\$HtosVz...	user	2025-05-17 22:34:09	NULL	NULL	NULL
	16	Kurt	Balais	cbalais@sfsu.edu	\$2b\$10\$ohiZ25...	user	2025-05-19 16:19:59	NULL	hi	user_16_...
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid				Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
	id	user_id	title	description	price	category	image_url	status	created_at	preferred_	item_c
▶	1	1	Acoustic Guitar	A gently used acous...	89.99	music	acoustic...	approved	2025-04-0...	NULL	NULL
	2	1	Harry Potter and the Sorcerer's Stone	First book in the Har...	19.99	book	harrypot...	approved	2025-04-0...	NULL	NULL
	3	1	Office Chair	Ergonomic office cha...	129.99	furniture	office_c...	approved	2025-04-0...	NULL	NULL
	4	1	Wireless Headphones	Noise-cancelling ove...	59.99	electronics	headpho...	approved	2025-04-0...	NULL	NULL
	5	1	Denim Jacket	Stylish blue denim ja...	39.99	clothing	denim_j...	approved	2025-04-0...	NULL	NULL
	7	1	Inception (Blu-ray)	Christopher Nolan's ...	14.99	movies	Inceptio...	approved	2025-04-0...	NULL	NULL
	8	1	To Kill a Mockingbird	A classic novel by H...	12.50	book	TKAMBo...	approved	2025-04-0...	NULL	NULL
	9	1	To Kill a Mockingbird (1962 Film)	Classic film adaptati...	9.99	movies	TKAMBM...	approved	2025-04-0...	NULL	NULL
	10	1	Apple AirPods Pro (2nd Gen)	Active noise cancella...	149.99	electronics	airpods2...	approved	2025-04-0...	NULL	NULL
	11	1	tralalao goes Tralalla	Super Cool Card ver...	2000...	other	1746243...	pending	2025-05-0...	NULL	NULL
	12	1	Adoring Fan	I dont want him any...	0.00	other	1746244...	pending	2025-05-0...	NULL	NULL
	13	1	Random Item Listing	I dont want this item...	2.00	other	1746340...	pending	2025-05-0...	NULL	NULL
	14	1	Large Image Test	This test shows wha...	1.99	other	1746401...	pending	2025-05-0...	NULL	NULL
	15	1	Little Shark Fella	I dont want him	1.99	other	1746735...	pending	2025-05-0...	NULL	NULL
	18	6	session test 2	just checking sessions	23.00	other	1746844...	pending	2025-05-1...	SFSU	used
	20	9	ipad pro 11 inch	2021 iPad Pro 11 inc...	650.00	electronics	1747024...	pending	2025-05-1...	Library L...	used
	21	9	Dreo Tower Fan	Dreo Tower Fan wit...	35.00	electronics	1747026...	pending	2025-05-1...	Library L...	used
	22	9	Don't Eat The Marshmallow Yet!	Don't Eat the Marsh...	15.00	book	1747027...	pending	2025-05-1...	Library L...	new
	30	5	Desk and Chair	Dont want it anymore	2.00	furniture	1747253...	approved	2025-05-1...	Universi...	used
	32	5	Old Backpack	Old backpack that i ...	22.22	other	1747254...	approved	2025-05-1...	Library L...	used
	33	8	Logitech G PRO X SUPERLIGHT 2 Wi...	Logitech G PRO X SU...	99.99	electronics	1747388...	approved	2025-05-1...	SFSU	used
	34	7	Older Macbook	Old MacBook that I ...	99.99	electronics	1747610...	sold	2025-05-1...	Universi...	used
	36	9	extension cord	white extension cord	15.00	electronics	1747621...	approved	2025-05-1...	Library L...	used
	38	8	Dell Pro 24 Plus Monitor	new dell monitor	199.99	electronics	1747628...	approved	2025-05-1...	Library L...	new
	39	8	Set of Four Trader Joe's Mini Canva...	Set of Four Trader J...	77.99	other	1747629...	approved	2025-05-1...	Universi...	new
	40	7	Horrible Math Textbook	Ill pay you to get rid...	1.00	book	1747630...	approved	2025-05-1...	SFSU	used
	41	11	bose headphones	used a couple of times	150.00	electronics	1747641...	pending	2025-05-1...	Library L...	used
	42	16	Scuderia Ferrari	Pretending to be a s...	1.00	other	1747671...	pending	2025-05-1...	Universi...	used
	43	16	Sum book	415 book	15.00	book	1747704...	sold	2025-05-2...	Universi...	used
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ERD: (NOTE: Messages and admin_review_log currently not used)



V. Github Organization:

Our GitHub master is Luis.

Branches: main, backend, frontend, development, frontendReact-old

Everyone had access to our main branch, but Jonathan and Kurt mostly used it. Other people will do their assignment in other branches first, then it will be merged into the main branch.

[GitHub home page](#)

Our GitHub home page:

The screenshot shows the GitHub interface for the repository 'csc648-fa25-0104-team08'. The repository is private and was forked from 'CSC-648-SFSU/csc-648-sfsu-2025-fall-souza-01-csc648-base-repo'. It has 5 branches and 0 tags. The current branch is 'main', which is 172 commits ahead of the upstream 'main'. The repository has 173 commits and was last updated yesterday. The repository description is 'csc-648-sfsu-2025-fall-souza-01-csc648-base-repo created by GitHub Classroom'. The repository has 0 stars, 0 forks, and 0 releases. The repository is owned by 'nity0122'.

The repository structure is as follows:

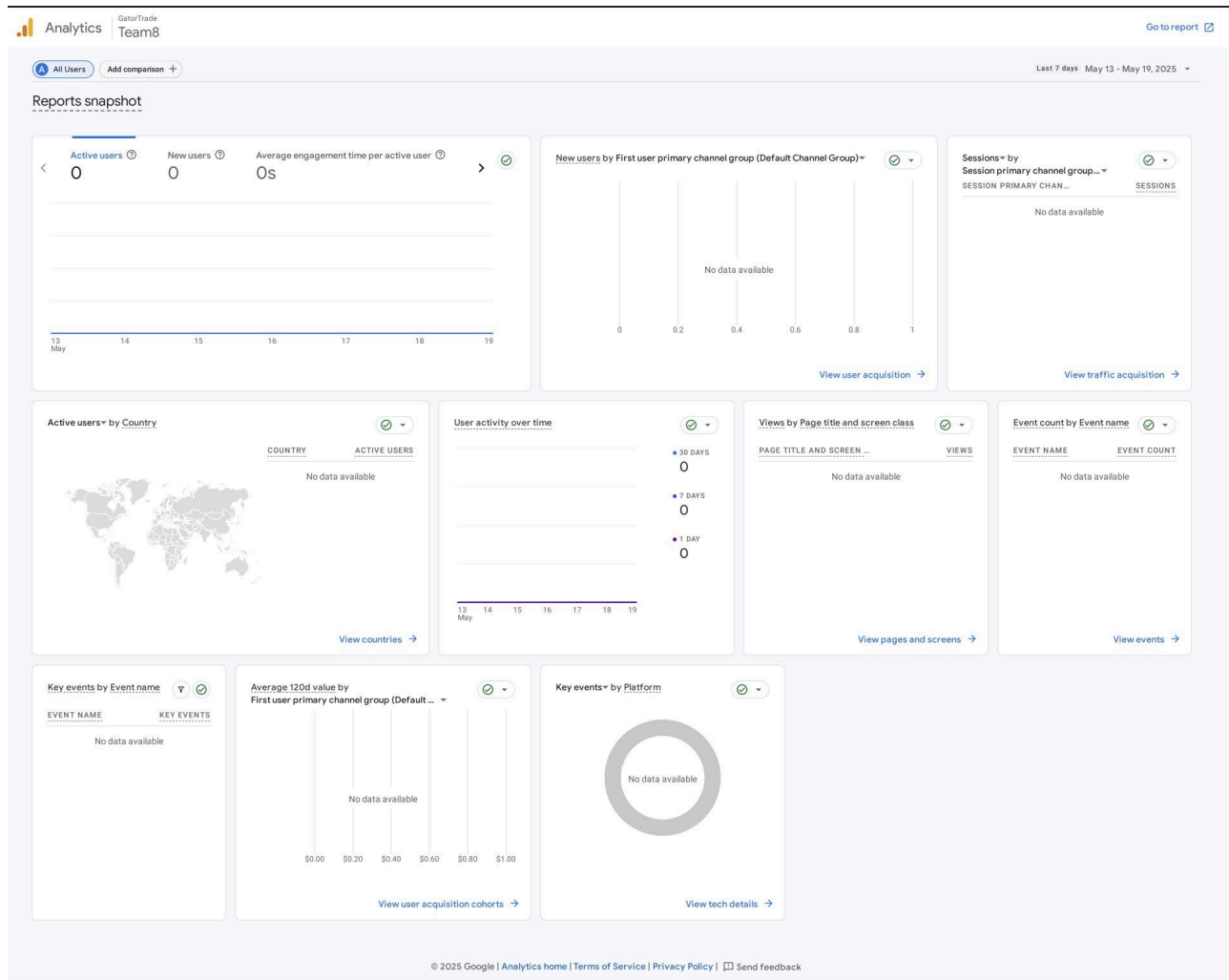
File	Last commit message	Last commit date
Milestones	added m2 part 1 revisions	last month
application	merging to main	2 days ago
credentials	updated IP address	2 weeks ago
.gitignore	removed node_modules, remove cors, changed backend call...	2 weeks ago
LICENSE	Initial commit	3 months ago
README.md	Update username.md	yesterday

The repository is viewed on the 'application' branch. The repository is 172 commits ahead of the upstream 'main'. The repository has 173 commits and was last updated yesterday. The repository description is 'csc-648-sfsu-2025-fall-souza-01-csc648-base-repo created by GitHub Classroom'. The repository has 0 stars, 0 forks, and 0 releases. The repository is owned by 'nity0122'.

The repository structure is as follows:

Name	Last commit message	Last commit date
..		
client	merging to main	2 days ago
controllers	add search word limit	3 days ago
images	edit fsu map	last week
images_profiles	refactored some code, fixed searching to work with Luis filtering, ma...	last week
models	updated db.js to use connection pool and made searchController async	last month
routes	fixed profile json bug, added team's pages, fixed some footer placeme...	3 days ago
.env	fixed profile json bug, added team's pages, fixed some footer placeme...	3 days ago
.gitignore	removed public folder	3 weeks ago
README.md	Initial commit	3 months ago

VI. Google Analytics:



Unfortunately, Google Analytics did not have enough data to show since it takes time for it to update, and Google Analytics was done late in the process.

VII. Project Management:

We used **Discord** as our primary project management and coordination tool. Rather than relying on formal task boards like Trello, we assigned and tracked tasks directly through structured team messages. Each member was given specific responsibilities related to frontend, backend, authentication, and deployment, with deadlines and coordination handled via chat.

Our Discord messages show how we divided work by component (e.g., registration, login, sessions), assigned teammates to tasks, provided technical resources (such as DB connection instructions), and gave reminders about branch usage. This setup worked effectively for our team's workflow, allowing for real-time updates and support.



Kaps 4/30/2025 8:34 PM

Registration Page:

Nathan:

- In the registration.jsx, change the red `***` so it turns green when the person enters the correct information. Remember that any validation in jsx file needs to be cosmetic only, the real check is going to be in the auth.js in the routes folder.
- Make it so the footer in registration is at the bottom of the page. If you figure out how to do it with reg, see if you can change the login page too.

Arthur:

- Work with Nathan to make a call to the backend auth.js to authenticate the inputs of user inputs and respond in two ways.
- If registration is successful, route to the login
- If registration is unsuccessful, print that "All fields are required", the user should be able to use the `***` to see what they are missing.
- If it goes through, check the database to see if the registration information is stored in the user table. If you don't know how to do this, let me know, and I can check for you real quick or teach you how to check.

Login Page:

Kurt:

- Implements sessions through Express, according to the professor, there's an easy and automatic way to do this.
- If you have extra time, start planning how we are going to do messages. For the most part, you'll be in charge of it next week.
- Help out if any of our front-end leads need help connecting to the backend.

Luis:

- You'll be working on getting login to work with the backend code. In auth.js, you can check whether there's an account with those credentials or not. Since registration is being worked at the same time, you'll have to test by creating a user in the DB, let me know if you need help with this. Respond in two ways after they try a login.
- If login is successful, reroute to the home page and initiate the session
- If login is unsuccessful, refresh the login page and show that the login is unsuccessful.
- If sessions and logins are working, figure out how to change the header depending on whether the user is logged in or not. Show sign in if they are not logged in, show profile if they are.

I'll handle creating a post on my own, shouldn't be too bad since I've done it before. With this, I can also experiment on how to lower the resolution of the images based on old thumbnail code I did before. Try to have this done by Sunday, by then I'll also start changing the little things, like having actual listings on the home page. Next week, we'll work on viewing posts, messaging, and the profile page, as well as milestone 4. As always, let me know if you need help with anything. Good luck team, we got this!!!

Heres instructions on how to connection to the db through SSH:

<https://docs.google.com/document/d/1ueLqgdRwGi9jOPHgZ7Ua57VJQiK9TMQ7LLwWRbLnCA/edit?usp=sharing>

VIII. Self-assessments and Contributions:

(Please zoom in if possible)

Chia-Tung Liu

self-assessment and contributions

CL

Chia-Tung Liu

收件者: Luis Alberto Moreno; Christian Kurt Bolanos Balais; Nathan Lee; Jonathan Rodrigo Carbajal

週二 2025/5/20 下午 04:11

😊

↩ 回覆

↩ 全部回覆

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a. I'm the front-end lead and designed the member page and all the webpages on paper for MS2. I have implemented three React pages: the Create Item page, Location Confirmation page, and Item page. I also connected the sign-up form to the back-end, integrated the Profile page to display user information from the back-end, and added several enhancements to the sign-up functionality. Additionally, I tested the search feature and created the QA report.

b. 23 (Mob1001+ArthurLiu24) I didn't commit that much, but I contributed through design work and testing. In React, I handled a larger portion of the workload in fewer commits, which might make it look like I contributed less based on commit count alone.

c. For me, learning React for the first time was a big challenge. I initially thought it would be difficult to pick up in a short amount of time. However, the good thing about React is that it's widely used, so there are plenty of online resources available to help me quickly understand the basics. ChatGPT also helped by explaining the parts I wasn't sure about, which made the learning process smoother.

d. I think GenAI helped me with brainstorming design ideas, such as deciding which functions should go on which pages and providing basic examples while I was designing. It also supported my learning of React by explaining code I found online and helping me debug issues in the code I wrote myself.

e. After this project, I've improved my skills in React and gained a better understanding of databases. I feel more confident now and believe I'll be more effective in these areas when working on future software engineering projects.

f. I think our team lead did a great job by assigning tasks based on each member's strengths, running efficient meetings, responding quickly to any questions I had, and—with the back-end lead Kurt—taking on a lot of work to ensure the project ran smoothly.

↩ 回覆

↩ 全部回覆

➡ 轉寄

Christian Kurt Balais

C

Christian Kurt Bolanos Balais

To: Jonathan Rodrigo Carbajal; Nathan Lee; Chia-Tung Liu; Luis Alberto Moreno

Tue 5/20/2025 3:01 PM

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a) Contributions to Team Project and Teamwork

Role: Backend Lead

- Set up and configured AWS EC2 instance and deployed the full stack web application (Node.js, MySQL, NGINX).
- Designed and implemented Express backend with REST API endpoints for user authentication, item posting, search, image upload, and messaging.
- Created and documented MySQL database schema for listings, users, messages, and images.
- Integrated React frontend with backend endpoints and enabled image uploads to server-side folder.
- Built out the search filtering functionality with category and keyword support using SQL LIKE and conditional queries.

b) GitHub Dev Submissions

Total commits to main branch: 54

Most commits were related to setting up server-side routes, debugging deployment issues, and implementing key backend features. A few commits were also for urgent fixes pushed from the server since I have been doing most of my work on the server itself.

c) Challenges Encountered

One major challenge I faced was configuring the deployment pipeline between React, Express, MySQL, and NGINX on the EC2 server. There were multiple issues with file permissions, routing between frontend and backend, and environment variables not loading correctly. I also encountered problems with npm install freezing during deployment, which led to exploring SCP and alternative installation strategies.

d) Experience with GenAI (ChatGPT / Copilot)

I used GenAI (ChatGPT) extensively throughout the project for backend development, and troubleshooting deployment errors (e.g. Express route handlers, MySQL queries, multer config). It was most useful for quick syntax generation and debugging server errors like ECONNREFUSED or nginx misconfigurations. I also used it to help generate the search filtering logic and QA test plans. Overall, GenAI saved a lot of time and helped me work more independently without needing to Google every error.

Best,

Christian Kurt Balais

Treasurer | ACM @ SFSU

Treasurer | SF Hacks

Jonathan Carbajal

Overall Feedback From Team Lead (CSC-648 Self-Assessment)

JC

Jonathan Rodrigo Carbajal

To:

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a. Contributions to the Team:

Role: Team Lead

As the team lead, it was my job to assign tasks, oversee the quality of the project, and communicate with the professor/CTO when turning in milestones.

Assured Milestones were finished and pushed into the repo

Created React files for the project and taught the team the basics

Connected React with the backend Express

Translate HTML files (like login and Register) to React

Added Image support and fixed overall CSS and UI responsiveness

Connected Create-post, view-post, and home page to the backend.

Added loading screens when loading new pages to minimize jitter

Added the messageHub to the profile page, corrected the API call to the backend

Be the bridge between the frontend team and the backend Lead

Created a GatorTrade email and developed a system where users get automated messages

Be the team's full-stack developer

Fixed many, many bugs

Clean the repo when it gets messy

Answer every question the team brought up

Organized everything, assigned the tasks after meetings, and made sure everything worked in the end.

b. GitHub Commits:

Total Number of Commits: 42

My commits were for almost anything, from turning in the milestones, fixing bugs, adding features, refactoring code, etc. You name an objective, and I probably did it.

c. Main Challenges

As the team lead, one of my main challenges stemmed from not planning the project design in advance. Much of our development process involved pivoting in response to new information or evolving directions from the professor. Additionally, none of us, including myself, were familiar with React, which made it difficult to get up to speed within a limited timeframe. I had to quickly learn how to access the Linux server, connect the frontend with the backend API, and use MySQL Workbench so that I could guide the rest of the team effectively. It was difficult to adapt, but not impossible.

d. GenAI Usage

Since this was my first time working with many of these technologies and languages, I relied on GenAI to help me learn and write code more quickly and efficiently. I believe I would have had a harder time without it, because the time it saved allowed me to focus more on the challenging parts of the project. It was like having a personalized Google search that could walk me through different approaches to solving problems in real time. For this project, it was extremely useful. That said, I'd be more hesitant to use it when I have more time, since I believe struggling through problems is an important part of the learning process.

e. Better Next Time:

With the information I learned this semester, building a better-looking website with cooler and "future-proof" features is a lot more plausible. I got extremely lucky with my team, as they would do their assignments on time, but using a project management system might allow for might make assignment management a lot more streamlined. For the later stages of the project, I started building several features on the spot since I had extra time. Looking back, I realize it would have been better to involve my team more in tackling the harder parts and brainstorming additional features we could have added, even within the short timeline. Next time, I'll make collaboration a bigger focus during those final phases.

f. Information For Instructor:

After hearing rants from my friend about his team, I'm so so blessed to have a wonderful team like mine who listens, does their work on time, and never once has let me down. Even if the commits tell a different story, everyone here has done an amazing job with every task I assign them. They aren't afraid to ask questions and all learned what they needed to learn for this project, even if I didn't ask sometimes. If it's possible, please give my teammates as much extra credit as they can; they're the best team I've had in my entire academic career.

Luis Moreno:

648 Contribution

LM

Luis Alberto Moreno

To:

Jonathan Rodrigo Carbajal;

Nathan Lee;

Christian Kurt Bolanos Balais;

Chia-Tung Liu

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Contribution to the team project and teamwork

Role: GitHub master

I helped design the messaging hub and message pages. I also helped redesign some pages, like profiles and listings. Most of my work was front-end, but I also worked a little with the back end. I helped my team create login sessions and keep them secure with session-based authentication. I helped design the filtering system and connect it to the database for real-time filtering on the page. Also was able to help my groupmate with his github name not showing up on earlier commits.

GitHub contributions.

GitHub contributions

I made a total of 15 commits. I have low commits, but these commits contain a lot of changes to the files and contribute a lot to the project as a whole.

Challenges

The biggest challenge was learning React in a short time frame. I was familiar with Java script so the transition to react was not that bad. The second challenge came from login, trying to keep the sessions and get data from the database was a problem.

Use of Gen AI

I used Gen AI to help me debug my react code and help me debug the terminal whenever I had trouble launching the site on my local machine. It was most useful to learn React more efficiently without searching for tutorials on how React works. It was a boost in productivity.

What I would do differently next time

I would focus on the software stack earlier on. I should've learned to react earlier in the semester rather than developing while learning. That would allow me to focus more on debugging the code. +4

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Tung-Ying Lee:

Self-assessment

NL

Nathan Lee

To: Chia-Tung Liu; Christian Kurt Bolanos Balais; Jonathan Rodrigo Carbajal; Luis Alberto Moreno

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Tue 5/20/2025 11:29 AM

a. Contributions to the team project and teamwork (Role: Frontend Developer)

- Started with static HTML and CSS, then converted everything to React using JSX
- Designed the index (landing) page
- Built the login page and started the registration page
- Contributed ideas for improving the profile page
- Focused on UI layout, component structure, and page interactions
- Tested the search function and the messaging system to ensure instant updates without refreshing
- Wrote the peer code review
- Participated in team meetings and followed the team lead's guidance

b. GitHub Contributions

- Made approximately 32 commits across two GitHub accounts: nty0122 and unknown
- My GitHub commit count may seem low, but I spent significant time on front-end development, including component logic, styling, layout improvements, and debugging
- I preferred committing only when features were polished and tested

c. Main challenges

The hardest part of this project was transitioning from basic HTML/CSS to using React for the first time. Learning JSX syntax and understanding React's core concepts—like state, props, and component lifecycles—was challenging but rewarding.

d. Experience with GenAI

GenAI tools like ChatGPT helped me accelerate my learning of React. I used it to convert HTML into JSX, manage state and props, structure UI components, debug errors, and understand best practices. It greatly improved my productivity and gave me more confidence in my work.

e. What I would do better next time

Next time, I would use React from the beginning to avoid unnecessary rework and time spent converting static pages. I now understand the importance of starting with a modular component structure and pushing updates frequently with clear commit messages. I also learned how early prototyping and regular team check-ins improve overall workflow and help balance frontend-backend collaboration.

f. Anything important for instructors to know

This is my first time using React and MySQL. I focused entirely on front-end development, while the back-end work was handled by my teammates. My team leaders mentored me early on, and with their support, I learned how to use React and MySQL and became more confident working independently. I think our team lead did an amazing job assigning tasks based on each member's strengths, running efficient meetings, answering all my questions quickly, and taking on a lot of work along with Kurt, the backend lead, to keep the project running smoothly.